

The matching benefits of market thickness*

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Abstract

The ability of larger markets to mitigate the incentive problem created by private information has been the focus of a sizable economics literature. In contrast, the fact that thicker markets also reduce the double coincidence of wants problem has received little attention. Modeling thin markets as bilateral trade involving independent private values and thick markets as Walrasian markets with a continuum of traders, we analyze and quantify the *matching benefits* of market thickness. These benefits increase in the *nicheness* of a product, which we measure as the mass of values and costs outside an interval of overlapping support where there are positive gains from trade. For sufficient nicheness, profit-maximizing intermediaries operating thick markets outperform ex post efficient bilateral trade. However, with bilateral trade as an outside option, traders of niche products are most vulnerable to intermediaries' market power. Extensions consider fixed costs of operating thick markets and finitely thick markets.

Keywords: Bilateral trade, Walrasian markets, thick market monopoly, market design, niche products

JEL-Classification: D47, D82, L12

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1 Introduction

Market thickness is commonly perceived as benign and desirable. An aspect of market thickness that the economics literature has long recognized and emphasized is that increasing the number of traders alleviates what may be called the *incentive problem*. That is, increasing market thickness decreases the distortions in market performance associated with agents' private information. An aspect of market thickness that is independent of the incentive problem and that has received rather scant attention in the economics literature is that thicker markets mitigate what may be dubbed the “double coincidence of wants” problem: A trader who would trade in a thick market might not trade in a thin market because the agent(s) on the other side of the market end up having bad draw(s). Due to mismatches of this kind, a valuable trader is lost in the thin market. The ability of thicker markets to mitigate the double coincidence problem is what we refer to as the *matching benefits* of market thickness.¹

In this paper, we provide a framework that permits analyzing and quantifying the matching benefits of thick markets. Thin markets are captured as bilateral trade with independent private values and a Walrasian market involving a continuum of traders—with demand and supply given by the same distributions as in the bilateral trade setting—captures the benchmark of a perfectly thick, competitive market. While many parameters and questions may be of interest, our particular focus will be on how the matching benefits depend on the degree to which a product is *niche*, with a product being more niche if trade is less likely to occur under ex post efficient bilateral trade. Formally, we model the degree of nicheness using a truncation model according to which a mass μ of buyers' values and sellers' costs lies outside the interval of overlapping support where there are positive gains from trade. The larger the mass μ is, the more niche the product is. This model enables us to perform comparative statics while allowing for reasonably general distributions from which buyers and sellers draw their types.² This specification also means that the more niche a product

¹To get a first sense for why these matching benefits may matter, recall that in the bilateral trade problem of Myerson and Satterthwaite (1983) with uniformly distributed values and costs, the ratio of second-best social surplus (9/64) to social surplus under ex post efficiency (1/6) is 0.84. In contrast, the ratio of social surplus under ex post efficiency for bilateral trade to social surplus per trader pair in a Walrasian market with an equal mass of buyers and sellers and uniformly distributed values and costs (1/4) is 0.66. That is, the matching benefits of market thickness outweigh the benefits from eliminating the incentive problem associated with bilateral trade. Another way to illustrate the strength of the matching benefits of thick markets is the fact that a profit-maximizing intermediary that operates a thick market with uniformly distributed values and costs generates more social surplus (3/16) than ex post efficient bilateral trade.

²Alternatively, one could directly vary the distributions themselves in order to control the probability with which trade occurs under ex post efficiency in a bilateral setting. However, this requires imposing more specific restrictions on the distribution and how they vary with the nicheness parameter μ . As we show in Appendix B.5, under the assumptions we impose there this modeling choice leads to largely the same

is, the smaller is the quantity traded in the thick, Walrasian market. We simply refer to this quantity as the *Walrasian quantity*. The specification captures in a simple and tractable way the double coincidence of wants problem: The probability of trade under ex post efficient bilateral trade given μ is $(1 - \mu)^2$ times the probability of trade under ex post efficient bilateral trade without truncation, while the Walrasian quantity given μ is $1 - \mu$ times the Walrasian quantity without truncation.

With this framework at hand, we show the following. First, the matching benefits of thick markets—measured as the ratio of expected social surplus per trader pair in the thick Walrasian market to expected social surplus under ex post efficient bilateral trade—are increasing in the nicheness of a product. This has the perhaps counterintuitive implication that, measured in these relative terms, thick markets are particularly valuable when the Walrasian quantity is small. The reason is simply that a small Walrasian quantity means that the probability of trade in a thin market is also small. Thanks to the law of large numbers, a high-value buyer or a low-cost seller trade with probability one in a thick market but they only trade if the other agent also has a favorable draw in the bilateral trade setting. This reflects the double coincidence of wants problem of thin markets.

Second, we show that for sufficiently niche products, the matching benefits of thick markets are so large that the expected surplus for each trader exceeds social surplus under ex post efficient bilateral trade, even if the thick market is operated by a profit-maximizing intermediary rather than a benevolent Walrasian auctioneer. Indeed, we show that for sufficiently niche products, the ratio of social surplus per trader pair in a market operated by a profit-maximizing intermediary to ex post efficient bilateral trade is increasing in the degree of nicheness whenever there are at least two pairs of traders. Moreover, for sufficiently niche products and uniformly distributed values and costs, a profit-maximizing intermediary generates more social surplus per trader than ex post efficient bilateral trade, even with only three trader pairs. An implication of these comparisons to monopoly intermediaries in thick markets is that traders of niche products benefited the most (in relative terms) from the emergence of large online platforms in the digital age.³ That said, and as we show formally in an extension, if bilateral trade serves as an outside option that constrains the market power of thick market intermediaries, then niche traders are also the most at risk of being exploited by these intermediaries precisely because bilateral trade is so ineffective due to the double coincidence of wants problem.

Last, assuming that there is a fixed cost of establishing and operating a thick market,

conclusions as the baseline specification involving truncation.

³For example, in the early 2000s the number of book titles available on Amazon was twenty times that of a typical, traditional book store (see Brynjolfsson et al., 2003; Waldfogel, 2017).

we show that thick markets are either established for products of intermediate nicheness or for the least niche products, regardless of whether the decision is made by a benevolent social planner or a profit-maximizing intermediary. The thick market is never established for the most niche products simply because there are too few of these traders to make it worthwhile. Among other things, this thought experiment provides guidance for policy makers pondering what kinds of products and services they should establish a centralized exchange for. However, if there are economies of scope that allow buyers and sellers of niche products to trade in the same markets as buyers and sellers of mass products then the former traders benefit from the presence of the latter. This last case seems descriptive of many online platforms.

In sum, this paper draws attention to the matching benefits of thick markets. It provides a tractable framework to analyze and quantify these benefits. It shows that the matching benefits can easily be so large that (in terms of expected social surplus per trader) a thick market monopoly outperforms ex post efficient bilateral trade. It also shows that while traders of highly niche products benefit the most (in relative terms) from thick market monopolies, they are also the most exposed to their market power when bilateral trade serves as an outside option. If market thickness comes at a fixed cost, then thick markets tend to be established for products of intermediate nicheness or for all products below a threshold level of nicheness.

As such, this paper contributes to three strands of literature. First, as mentioned, the incentive problem (see, for example, Vickrey, 1961; Myerson and Satterthwaite, 1983) has received considerable attention in the double auctions and mechanism design literature. For example, Gresik and Satterthwaite (1989), Satterthwaite and Williams (1989), McAfee (1992), Rustichini et al. (1994), Satterthwaite and Williams (2002), Tatur (2005), Cripps and Swinkels (2006), Kojima and Yamashita (2017) and Loertscher and Mezzetti (2021) show that the incentive cost vanishes quickly as market size increases. Our paper contributes to this strand of literature by showing that the matching benefits of thick markets can be a first-order issue, thereby formalizing in a one-shot setting a property observed by Loertscher et al. (2022) in a dynamic setup. Indeed, using ex post efficient bilateral trade as a thin market benchmark rules out any incentive benefits from increasing market thickness. Yet, social surplus per trader pair can easily dwarf social surplus under bilateral trade. This is not meant to deny the importance of the incentive problem in thin markets. However, in light of the results from the double auctions literature and assuming that the incentive problem can be tackled along the way, it may be taken to suggest that market designers should focus on the matching benefits of market thickness. The tractability of our framework derives from two invariance results that relate to the truncation invariance of virtual type functions.

Among other things, this implies that the ratio of second-best social surplus divided by social surplus under ex post efficiency in bilateral trade is independent of the degree of nicheness. For example, for the case discussed in footnote 1, it is 0.84 for any μ . In contrast, the ratio of per trader pair social surplus under ex post efficient bilateral trade over that in a thick Walrasian market is $2/3$ times $1 - \mu$. Similarly, relative to an efficient thick market, the harm from a thick market monopoly does not vary with how niche a product is. This tractability is also what permits us to analyze the tradeoffs between market thickness and the fixed costs of establishing thick markets and to show that these are in many ways similar to those in the dynamic setting of Loertscher et al. (2022).

Second, the matching benefits of thick markets relate to Oliver Williamson’s observation that larger firms should outperform smaller ones because they can always replicate what the smaller ones do, and sometimes do better by *intervening selectively* (Williamson, 1985). Exactly the same logic drives the matching benefits of market thickness. Selective intervention in market making takes the form of matching traders who, in bilateral trade settings, would not be matched to each other. Selective intervention becomes increasingly more valuable as products become more niche. While the resulting increasing returns to scale have received scant attention in the theoretical literature,⁴ they are the driving force behind the emergence of online platforms in the digital age, where they are well documented. For example, according to Waldfogel (2017), digitalization has led to a “golden age of music, movies, books and television.” Anderson (2006) provides an early account of the benefits that accrue to consumers (and producers) from the access to the “long tail” that digitalization grants. He notes that “Netflix changed the economics of niches, and in so doing, reshaped our understanding about what people actually *want* to watch” and reports that for books, “Barnes & Noble found that the bottom 1.2 million titles represent just 1.7 percent of its instore sales, but a full 10 percent of its online (bn.com) sales.”⁵ Our paper contributes to this literature and the related, ongoing policy debates by showing that even thick market monopolies can outperform ex post efficient trade with regard to per trader producer and consumer surplus.

Last, with the extension that assumes that bilateral trade serve as an outside option to trading with a thick market monopoly, this paper also contributes to the literature in which

⁴A notable exception is Bulow and Klemperer (1996) who demonstrate the importance of increasing the number of buyers in one-sided settings by showing that a second-price auction with no reserve price but one additional bidder outperforms the optimal auction. Of course, market thickness and thinness also play an important role in the literature on financial markets, where they are typically related to an agent’s price impact in a continuous-time double auction (see e.g. Rostek and Weretka, 2008). In contrast, the definitions used here (and in Loertscher et al., 2022) are independent of the trading mechanism, implying that a market can be thin even if a dominant-strategy mechanism—such as the one from McAfee (1992)—is used.

⁵Brynjolfsson et al. (2003) estimate that Amazon’s book sales for titles ranked outside the top 100,000 accounted for 39.2% of total book sales of Amazon.

profit-maximizing intermediaries face a competing exchange such as Gehrig (1993), Spulber (2002), Rust and Hall (2003), and Loertscher and Niedermayer (2020).

The remainder of this paper is organized as follows. Section 2 introduces the setup. In Section 3 we derive our results concerning thick Walrasian markets and thick market monopolies. Section 4 analyzes market thickness for cases where establishing a thick market involves a fixed cost, and the implications of competing bilateral exchanges for market power. Section 5 concludes the paper. All proofs are provided in Appendix A and additional extensions and robustness checks can be found in Appendix B.

2 Setup

Throughout this paper we consider static market settings involving risk-neutral buyers and sellers with unit demand and unit capacity, respectively, whose utility functions are quasi-linear. We assume that each buyer is privately informed about the realization of its value v and each seller is privately informed about its realized cost c . Under these assumptions, optimal mechanisms are well-defined without restrictions on the contracting space. With the exception of the analysis in Section 4.2, the value of the outside option of not participating in the market is 0 for every agent.

We first introduce the notion of a *mass product*. In markets for a mass product buyers and sellers draw their types from distributions with a common support. Specifically, buyer values $v \in [0, 1]$ are distributed according to the distribution F with density f that has full support on $[0, 1]$. Similarly, seller costs $c \in [0, 1]$ are distributed according to the distribution G with density g that has full support on $[0, 1]$. For any $\alpha \in [0, 1]$, the *weighted virtual type functions* are then

$$\Phi_\alpha(v) := v - \alpha \frac{1 - F(v)}{f(v)} \quad \text{and} \quad \Gamma_\alpha(c) := c + \alpha \frac{G(c)}{g(c)}.$$

We also let $\Phi(v) := \Phi_1(v)$ and $\Gamma(c) := \Gamma_1(c)$ denote the *virtual type functions* that have weight of $\alpha = 1$ on revenue. To develop an intuitive understanding of these functions, consider first Φ and Γ . As observed by Mussa and Rosen (1978) and Bulow and Roberts (1989), these functions can be interpreted as marginal revenue and marginal (procurement) cost functions. To see this, notice first that $F^{-1}(1 - q)$ and $G^{-1}(q)$ can be viewed as the inverse demand and supply functions given quantity $q \in [0, 1]$, implying that revenue and procurement cost as a functions of quantity q are $H_F(q) := qF^{-1}(1 - q)$ and $H_G(q) := qG^{-1}(q)$. Consequently, the marginal revenue and marginal cost are $H'_F(q)|_{p=F^{-1}(1-q)} = \Phi(p)$ and $H'_G(q)|_{p=G^{-1}(q)} = \Gamma(p)$. Notice next that $\Phi_\alpha(v) = (1 - \alpha)v + \alpha\Phi(v)$ and $\Gamma_\alpha(c) =$

$(1 - \alpha)c + \alpha\Gamma(c)$. That is, the weighted virtual type functions are convex combinations of the true types v and c and the virtual types $\Phi(v)$ and $\Gamma(c)$, with weight α on the virtual types. Consequently, the weighted virtual type functions $\Phi_\alpha(v)$ and $\Gamma_\alpha(c)$ are the analogues to the marginal revenue and marginal cost functions derived from a Ramsey pricing problem in which the regulator's weight on revenue is α .⁶ For any $\alpha \in [0, 1]$, we let $\bar{\Phi}_\alpha(v)$ and $\bar{\Gamma}_\alpha(c)$ respectively denote the weighted *ironed* virtual valuation and cost functions.⁷ As we just saw, Φ and Γ represent marginal revenue and marginal cost under *market-clearing* pricing. However, in cases where one of these functions is not monotone (and market-clearing pricing is not necessarily optimal for both sides of the market) the ironed functions $\bar{\Phi}$ and $\bar{\Gamma}$ represent marginal revenue and marginal cost under the *optimal* mechanism.

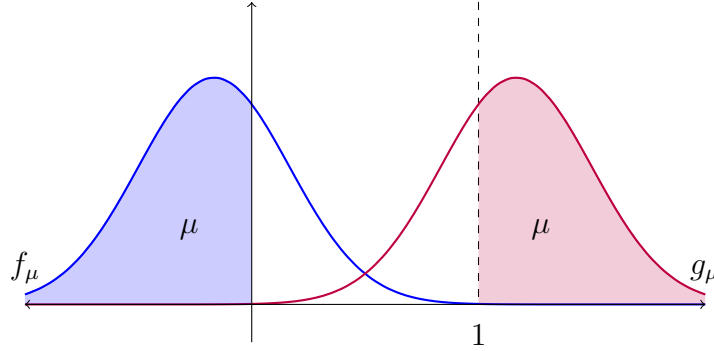


Figure 1: An illustration of the density functions f_μ and g_μ for a niche market.

In the market for a *niche product* with parameter $\mu \in [0, 1]$ we assume that buyer values $v \in [a_\mu, 1]$ with $a_\mu < 0$ are drawn from some distribution F_μ with density f_μ that has full support on $[a_\mu, 1]$ and seller costs $c \in [0, b_\mu]$ with $b_\mu > 1$ are drawn from some distribution G_μ with density g_μ that has full support on $[0, b_\mu]$. While these distributions may be arbitrary outside the interval $[0, 1]$ of overlapping support, on this interval these distributions are

⁶For more on Ramsey pricing, see, for example, Wilson (1993). Bulow and Roberts (1989) make the connection between the weighted virtual type functions used in mechanism design and Ramsey pricing while Myerson and Satterthwaite (1983) show that $\alpha = \lambda/(1 + \lambda)$ in the bilateral trade problem they study, where $\lambda > 0$ is the solution value of the Lagrange multiplier associated with the no-deficit constraint.

⁷The ironed virtual type functions are the marginal revenue and marginal cost functions associated with the concavification of the revenue function and the convexification of the procurement cost function, respectively; see, for example, Myerson (1981). Specifically, we let $\bar{H}_F$ denote the concavification of H_F and $\bar{H}_G$ denote the convexification of H_G . The ironed virtual type functions are then given by $\bar{\Phi}(v) = \bar{H}'_F(q)|_{q=1-F(v)}$ and $\bar{\Gamma}(c) = \bar{H}'_G(q)|_{q=G(c)}$, while the weighted ironed virtual type functions are given by $\bar{\Phi}_\alpha(v) = (1 - \alpha)v + \alpha\bar{\Phi}(v)$ and $\bar{\Gamma}_\alpha(c) = (1 - \alpha)c + \alpha\bar{\Gamma}(c)$. Here, the concavification (also known as the upper concave envelope) of a given function is the smallest concave function that is weakly larger than that function at every point in its domain. The convexification (or lower convex envelope) of a given function is the largest convex function that is weakly less than that function at every point in its domain.

restricted so that for all $\mu \in [0, 1)$, $v \in [0, 1]$ and $c \in [0, 1]$, we have

$$F_\mu(0) = 1 - G_\mu(1) = \mu, \quad \frac{F_\mu(v) - \mu}{1 - \mu} = F(v) \quad \text{and} \quad \frac{G_\mu(c)}{1 - \mu} = G(c).$$

That is, in a market for a niche product with parameter $\mu \in [0, 1)$, buyers have a value $v \in [a_\mu, 0)$ with probability $\mu \in [0, 1)$ and a value $v \in [0, 1]$ with probability $1 - \mu$. Conditional on $v \in [0, 1]$, buyers' values are distributed according to F . Similarly, sellers have a cost $c \in (1, b_\mu]$ with probability μ and a cost $c \in [0, 1]$ with probability $1 - \mu$. Conditional on $c \in [0, 1]$, sellers' costs are distributed according to G . See Figure 1 for an illustration. The following observation plays an important role in our analysis and contributes in no small parts to the model's tractability. We state it as a lemma since it seems a useful fact without claiming or believing that we are the first to make this observation:⁸

Lemma 1. *The weighted virtual type functions are truncation invariant. That is, for any $\alpha \in [0, 1]$, $\mu \in [0, 1)$, $v \in [0, 1]$ and $c \in [0, 1]$, we have*

$$\Phi_\alpha(v) = v - \alpha \frac{1 - F_\mu(v)}{f_\mu(v)} \quad \text{and} \quad \Gamma_\alpha(c) = c + \alpha \frac{G_\mu(c)}{g_\mu(c)}. \quad (1)$$

Most of the analysis studies two extreme cases that encompass both perfectly thin and perfectly thick markets. On the *thin* end of the spectrum we consider the *bilateral trade problem* of Myerson and Satterthwaite (1983) which involves exactly one buyer and one seller. While some of our results—such as Proposition 1—hold for any α , one benchmark for the bilateral trade problem is social surplus under the *second-best* mechanism, which maximizes expected gains from trade subject to agents' incentive compatibility and interim individual rationality constraints and a no-deficit constraint for the designer.⁹ By construction, the designer's expected profit under the second-best mechanism is 0. In contrast to the second-best mechanism, the allocation rule under the *first-best* (or ex post efficient) mechanism is to induce trade whenever $v \geq c$. On the *thick* end of the spectrum we consider markets involving a continuum of traders, with an equal mass of buyers and sellers on each side of the market. In this case we look at both the *thick Walrasian market*, in which there is a benevolent (arguably fictitious) auctioneer who organizes the exchange and quotes the

⁸While our analysis keeps the overlapping support of the distributions F and G fixed as $[0, 1]$, the property stated in (1) also holds for $v \in [\underline{v}, 1]$ and $c \in [0, \bar{c}]$ with $\underline{v} \in [0, 1)$ and $\bar{c} \in (0, 1]$ if the supports of F_μ and G_μ are $[0, 1]$ and $F(v) = \frac{F_\mu(v) - F_\mu(\underline{v})}{1 - F_\mu(\underline{v})}$ and $G(c) = \frac{G_\mu(c)}{G_\mu(\bar{c})}$.

⁹Formally, the second-best mechanism maximizes $\mathbb{E}[(v - c)Q(v, c)]$ over $Q : [0, 1]^2 \rightarrow [0, 1]$ subject to the aforementioned constraints. The solution to this problem is given by the allocation rule $Q^{\alpha^*}(v, c)$ that induces trade if and only if $\Phi_{\alpha^*}(v) \geq \bar{\Gamma}_{\alpha^*}(c)$, where $\alpha^* \in (0, 1)$ is the unique number such that the expected revenue of the designer is 0. That is, such that $\mathbb{E}[(\Phi(v) - \Gamma(c))Q^{\alpha^*}(v, c)] = 0$.

market clearing price, and the *thick market monopoly*, where the thick market is organized by a profit-maximizing monopoly that has zero costs.¹⁰

The setting introduced here provides a sharp illustration of the main insights of this paper. The results in Section 3 can be generalized to allow for asymmetric niche product parameters for buyers and sellers (see Appendix B.2) and to allow for the region of overlapping support between the type distributions to vary with μ (see Appendix B.3). Moreover, nicheness can be modeled in a variety of other ways without changing our main insights. For example, one could consider type distributions with unbounded support, where products become increasingly niche as probability mass is shifted towards the tails of the type distributions in an appropriate fashion (see Appendix B.4). One can also model niche products by parameterizing the type distributions so that the elasticity of demand and supply increases and the Walrasian quantity traded decreases as products become increasingly niche (see Appendix B.5). As shown in these appendices, the main insights carry over to these alternative specifications. Our baseline setup involving truncation is general insofar as we impose only mild restrictions on the distributions from which the agents draw their types. Its tractability without parameteric restrictions on the type distributions derives from the way the niche nature of products is modeled using a single parameter.

3 Of thin and thick markets, niche and mass products

For $\mu \in [0, 1]$, let $S_1^{FB}(\mu)$ and $S_1^{SB}(\mu)$ denote first-best and second-best welfare, respectively, in the bilateral trade setting.¹¹ Similarly, we let $S_\infty(\mu)$ denote welfare per trader pair in the thick Walrasian market. In this section we analyze the thick Walrasian market and the thick market monopoly by relating them to the bilateral trade problem. In particular, we will be interested in studying how $S_1^{FB}(\mu)$, $S_1^{SB}(\mu)$ and $S_\infty(\mu)$ and their ratios vary with μ . The mass product ratios

$$s_1 := \frac{S_1^{SB}(0)}{S_1^{FB}(0)} \quad \text{and} \quad s_\infty := \frac{S_\infty(0)}{S_1^{FB}(0)}$$

¹⁰See Aumann (1964) for a formalization of competitive equilibrium and its relation to the core with a continuum of traders. Implicitly or explicitly, the theory of monopoly and oligopoly pricing dating back to Cournot (1838) has rested on the assumption of a continuum of price-taking agents on sides of the market without power.

¹¹As shown by Myerson and Satterthwaite (1983), first-best trade is not possible when the agents are privately informed about their values and costs, their interim incentive compatibility and individual rationality constraints have to be satisfied and the mechanism must not run a deficit. Our assumptions imply that $0 < w < 1$ and that in the bilateral setting the probability of trade under ex post efficiency is less than 1.

with $0 < s_1 < 1 < s_\infty$ provide a natural benchmark.¹² We will also study how consumer surplus per buyer, producer surplus per seller, and social surplus per trader pair vary with μ , as well as how they compare to $S_1^{FB}(\mu)$ when the thick market is organized by a profit-maximizing monopoly.

3.1 Bilateral trade as model of thin markets

We begin by analyzing the bilateral trade problem. Since we have $f_\mu(v) = (1 - \mu)f(v)$ and $g_\mu(c) = (1 - \mu)g(c)$ for all $v \in [0, 1]$ and $c \in [0, 1]$, the probability of trade under ex post efficiency given μ is $(1 - \mu)^2$ times the probability of trade under ex post efficiency when $\mu = 0$.¹³ This captures in a nutshell the essence of the double coincidence of wants problem. Further, as noted, conditional on having both $v \in [0, 1]$ and $c \in [0, 1]$, agents' type distributions (and hence agents' virtual type functions) do not vary with the parameter μ . Consequently, the first-best mechanism and second-best mechanisms do not depend on μ and we have

$$S_1^{FB}(\mu) = (1 - \mu)^2 S_1^{FB}(0) \quad \text{and} \quad S_1^{SB}(\mu) = (1 - \mu)^2 S_1^{SB}(0), \quad (2)$$

where $(1 - \mu)^2$ is the probability that $v \in [0, 1]$ and $c \in [0, 1]$. More generally, for any $\alpha \in [0, 1]$ we can consider the mechanism that maximizes a convex combination of social welfare and designer profit, where α is the weight on the designer's profit. Letting $S_1^\alpha(\mu)$ denote welfare under this mechanism, we have the following proposition.

Proposition 1. *For any $\alpha \in [0, 1]$ and $\mu \in [0, 1)$, we have $S_1^\alpha(\mu) = (1 - \mu)^2 S_1^\alpha(0)$, where $S_1^\alpha(0) = \mathbb{E}[(v - c)Q^\alpha(v, c)]$ and $Q^\alpha(v, c) = \mathbb{1}(\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c))$. Consequently, the ratio $\frac{S_1^\alpha(\mu)}{S_1^{FB}(\mu)}$ does not vary with μ .*

The class of mechanisms considered in Proposition 1 is equivalent to the class of mechanisms that implement Ramsey pricing.¹⁴ These mechanisms trace out a frontier characterizing the optimal tradeoff between revenue and welfare, encompassing welfare maximization (the first-best mechanism), welfare maximization subject to budget balance (the second-best mechanism) and profit maximization.

¹²The impossibility theorem of Myerson and Satterthwaite (1983) implies $s_1 < 1$. Because $0 < w < 1$, there are matching benefits, that is, $s_\infty > 1$.

¹³The probability of trade under ex post efficiency is $\int_0^1 \int_0^v g_\mu(c) f_\mu(v) dc dv = (1 - \mu)^2 \int_0^1 \int_0^v g(c) f(v) dc dv$.

¹⁴Under Ramsey pricing social welfare is maximized subject to the constraint that the mechanism generates profit of at least K . The second-best mechanism corresponds to Ramsey pricing with $K = 0$. This class of mechanisms is also equivalent to the class of mechanisms that maximize profit subject to the constraint that some minimal level of welfare is achieved. See Wilson (1993) and Bulow and Roberts (1989).

In some applications, other mechanisms, such as the trade sacrifice mechanism of McAfee (1992), may be more practical. With that in mind, we have the following extension of Proposition 1 to a posted-price mechanism (Hagerty and Rogerson, 1987), which is what the trade sacrifice mechanism of McAfee (1992) specializes to when there is only one buyer and one seller. Letting $S_1^{HR}(\mu)$ denote Hagerty-Rogerson welfare in the bilateral trade problem, we have

$$S_1^{HR}(\mu) = (1 - \mu)^2(1 - F(p^*))G(p^*)\mathbb{E}[v - c|v \geq p^*, c \leq p^*] = (1 - \mu)^2 S_1^{HR}(0),$$

where $p^* = \arg \max_{p \in [0,1]} (1 - F(p))G(p)\mathbb{E}[v - c|v \geq p, c \leq p]$. This shows that the ratio $\frac{S_1^{HR}(\mu)}{S_1^{FB}(\mu)}$ does not vary with μ either.

3.2 Thick Walrasian markets

We now analyze thick markets involving a continuum of traders, starting with thick Walrasian markets. For this setting, the market clearing price p is independent of μ and such that $1 - F(p) = G(p)$. We also denote by

$$w = 1 - F(p) = G(p) \tag{3}$$

the *Walrasian quantity* in a market for mass products (strictly speaking, the probability that any given agent trades in an efficient thick market, conditional on its type being between 0 and 1). Since $1 - F_\mu(p) = (1 - \mu)(1 - F(p))$ and $G_\mu(p) = (1 - \mu)G(p)$, the Walrasian quantity given the niche parameter μ is simply $(1 - \mu)w$. Welfare per trader pair in the thick Walrasian market is then

$$S_\infty(\mu) = w(1 - \mu)\mathbb{E}[v - c|v \geq p, c \leq p] = (1 - \mu)S_\infty(0).$$

This setup and the role of the niche product parameter μ in thick Walrasian markets is illustrated in Figure 2.

Combining all of this with our analysis from the previous section, we have the following proposition.

Proposition 2. *For any $\mu \in [0, 1)$ we have $\frac{S_\infty(\mu)}{S_1^{FB}(\mu)} = \frac{s_\infty}{1 - \mu} > 1$ and hence $\lim_{\mu \rightarrow 1} \frac{S_\infty(\mu)}{S_1^{FB}(\mu)} = \infty$.*

Propositions 1 and 2 provide a formalization of the importance of avoiding the double coincidence of wants problem for niche products. As is well known from the double auctions literature, the incentive cost of small markets vanishes quickly as the size of a market increases. Since the second-best mechanism arises as a special case of Proposition 1, this

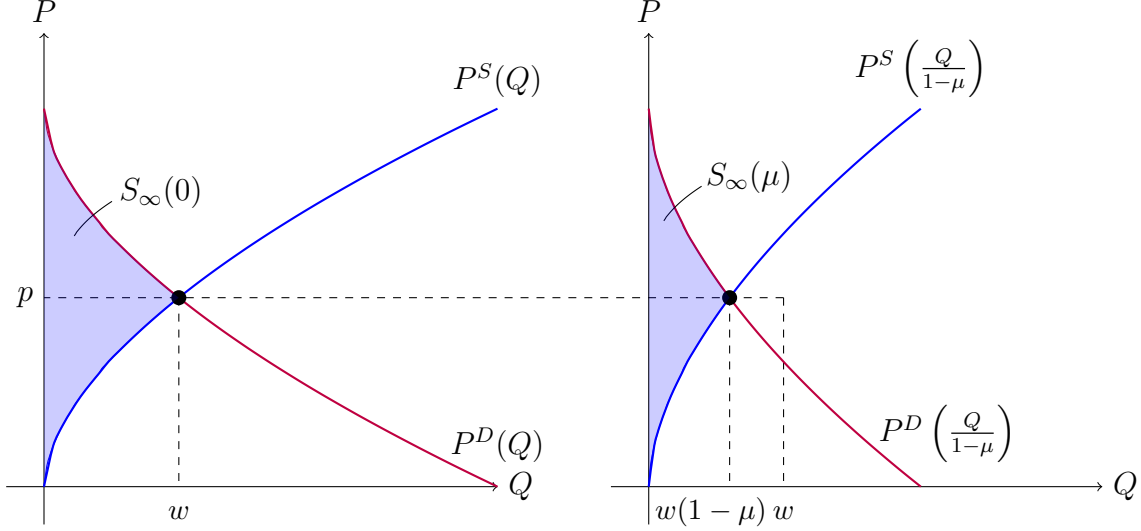


Figure 2: The thick Walrasian market for a mass product and for a niche product.

proposition shows that the ratio $\frac{S_1^{SB}(\mu)}{S_1^{FB}(\mu)}$, which bounds the incentive cost of small markets, does not depend on the niche market parameter μ . In contrast, the relative benefits of thick markets, captured by the ratio $\frac{S_\infty(\mu)}{S_1^{FB}(\mu)}$, increase in μ at rate $(1 - \mu)^{-1}$. Intuitively, thin niche markets suffer from a severe double coincidence of wants problem. First-best welfare in the bilateral trade setting is proportional to $(1 - \mu)^2$ because any given agent with a type that lies within the interval $[0, 1]$ is only matched with another such agent with probability $1 - \mu$. In the thick Walrasian market welfare per trader pair is proportional to $1 - \mu$ and any buyer with value $v \geq p$ and any seller with cost $c \leq p$ trades with probability 1.

3.3 Thick market monopoly compared to efficient bilateral trade

We now consider a thick market with niche product parameter μ that is operated by a monopoly intermediary that maximizes its profits. We let $CS_\infty^M(\mu)$, $PS_\infty^M(\mu)$ and $S_\infty^M(\mu)$ denote surplus per buyer, surplus per seller and total surplus per trader pair, respectively. In this case it is useful to distinguish whether or not the functions Φ and Γ are increasing. When these functions are increasing, customarily referred to as the *regular* case, this implies that the monopolist's marginal revenue and marginal cost functions are monotone (and respectively decreasing and increasing). The optimal mechanism under a thick market monopoly then involves posting prices p_B (for buyers) and p_S (for sellers) on both sides of the market. These prices equate marginal revenue and marginal cost and balance the quantity traded. That is, they satisfy

$$\Phi(p_B) = \Gamma(p_S) \quad \text{and} \quad 1 - F(p_B) = G(p_S). \quad (4)$$

To see this, notice that if the monopoly trades the quantity $q \in [0, 1]$ at market clearing prices $p_B = F^{-1}(1 - q)$ and $p_S = G^{-1}(q)$, its profit maximization problem is

$$\max_{q \in [0, 1]} (F^{-1}(1 - q) - G^{-1}(q))q, \quad (5)$$

which yields the first-order condition

$$\Phi(F^{-1}(1 - q)) - \Gamma(G^{-1}(q)) = \Phi(p_B) - \Gamma(p_S) = 0.$$

The second-order condition is satisfied if $\Phi(v)$ and $\Gamma(c)$ are increasing, proving that the first-order condition characterizes a maximum and hence that the prices in (4) are optimal and unique. Moreover, they are independent of μ and such that $0 < p_S < p_B < 1$. Let

$$m = G(p_S) = 1 - F(p_B) \quad (6)$$

denote the probability that any given buyer or seller trades under the thick market monopoly, conditional on having a value or cost between 0 and 1. The unconditional probability of trading is then $m(1 - \mu)$. Accordingly, in the regular case, the surplus per buyer and the surplus per seller under the thick market monopoly are

$$CS_{\infty}^M(\mu) = m(1 - \mu)\mathbb{E}[v - p_B | v \geq p_B] \quad \text{and} \quad PS_{\infty}^M(\mu) = m(1 - \mu)\mathbb{E}[p_S - c | c \leq p_S].$$

Moreover, the monopoly's profit per trader pair is $m(1 - \mu)(p_B - p_S)$ and consequently total surplus per trader pair is given by

$$S_{\infty}^M(\mu) = CS_{\infty}^M(\mu) + PS_{\infty}^M(\mu) + m(1 - \mu)(p_B - p_S).$$

More generally, if the type distributions are not restricted to be regular, we can apply the analysis of Loertscher and Muir (2022) to obtain similar expressions for consumer surplus, producer surplus and the monopoly's profit per trader pair.

An interesting and important question is whether, relative to a thick Walrasian market, a thick market monopoly is more harmful for consumers and producers with niche or with mass products. As we show next, the relative harm from monopolies that intermediate a thick market does not depend on the nature of the product. Letting $CS_{\infty}(\mu)$ and $PS_{\infty}(\mu)$ respectively denote consumer surplus per buyer and producer surplus per seller in a thick

Walrasian market we have

$$CS_\infty(\mu) = w(1 - \mu)\mathbb{E}[v - p|v \geq p] \quad \text{and} \quad PS_\infty(\mu) = w(1 - \mu)\mathbb{E}[p - c|c \leq p].$$

Putting all of this together yields the following proposition.

Proposition 3. *For the thick market monopoly, there exists a $\hat{\mu} < 1$ such that for all $\mu > \hat{\mu}$,*

$$CS_\infty^M(\mu) > S_1^{FB}(\mu) \quad \text{and} \quad PS_\infty^M(\mu) > S_1^{FB}(\mu).$$

Moreover, $\lim_{\mu \rightarrow 1} \frac{CS_\infty^M(\mu)}{S_1^{FB}(\mu)} = \lim_{\mu \rightarrow 1} \frac{PS_\infty^M(\mu)}{S_1^{FB}(\mu)} = \infty$ and the ratios $\frac{CS_\infty^M(\mu)}{CS_\infty(\mu)}$, $\frac{PS_\infty^M(\mu)}{PS_\infty(\mu)}$ and $\frac{S_\infty^M(\mu)}{S_\infty(\mu)}$ are finite and independent of μ for $\mu \in [0, 1)$.

In markets for products that are sufficiently “niche” both consumer surplus per buyer and producer surplus per seller under a thick market monopoly exceed total surplus under ex post efficient bilateral trade, and as μ approaches 1, the ratio of consumer surplus per buyer and producer surplus per seller under the thick market monopoly over first-best welfare in bilateral trade diverge to infinity. Of course, because social surplus is larger than the sum of consumer and producer surplus, Proposition 3 implies that social surplus per trader pair under the thick market monopoly over first-best welfare in the bilateral trade problem also diverge to infinity as μ approaches 1. However, as Figure 3 and Example 1 below show, thick market monopolies can easily outperform ex post efficient bilateral for values of μ that are well below 1 (see also Section 3.4, which provides conditions under which markets operated by a profit-maximizing intermediary outperform bilateral trade away from perfect thickness).

Proposition 3 also states an invariance result for thick markets: that the severity of monopoly harm does not vary with μ . This parallels the invariance result for bilateral trade as stated in Proposition 3, which shows that the severity of the incentive problem for thin products is independent of the nature of the product. Like Proposition 3, the results stated in Proposition 3 are driven by the truncation invariance of the virtual type functions.

Proposition 3 has two potentially important policy implications. First, it means that the justification for governments to intervene or not to intervene in thick monopoly markets is the same for niche and for mass products.¹⁵ Second, in combination with Proposition 3, it implies that for μ sufficiently large, the consumer and producer benefits from thick market monopolies relative to thin markets are first order, while the harm from monopolies relative to an efficient thick market is second order.

¹⁵See, however, Section 4.2, which shows that competing bilateral exchanges are more effective in curbing the market power of thick market monopolies for mass products than for niche products.

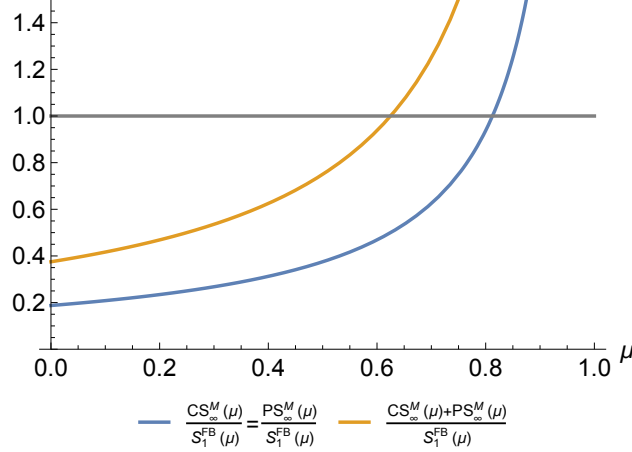


Figure 3: The ratios $CS_{\infty}^M(\mu)/S_1^{FB}(\mu) = PS_{\infty}^M(\mu)/S_1^{FB}(\mu)$ (blue) and $(CS_{\infty}^M(\mu) + PS_{\infty}^M(\mu))/S_1^{FB}(\mu)$ (orange) as functions of μ for F and G uniform.

The following example illustrates the results of Proposition 3 when buyer values and seller costs are uniformly distributed.

Example 1. *As an illustration, assume that F and G are uniform. Then $p_B = 3/4$ and $p_S = 1/4$, implying $m = 1/4$ and $CS_{\infty}^M(\mu) = PS_{\infty}^M(\mu) = (1 - \mu)/32$ while the market maker's profit is $m(1 - \mu)(p_B - p_S) = (1 - \mu)/8$. We also have $CS_{\infty}(\mu) = PS_{\infty}(\mu) = (1 - \mu)/8$ and $S_{\infty}(\mu) = (1 - \mu)/4$, which implies that $CS_{\infty}^M(\mu)/CS_{\infty}(\mu) = PS_{\infty}^M(\mu)/PS_{\infty}(\mu) = 1/4$ and $S_{\infty}^M(\mu)/S_{\infty}(\mu) = 3/4$. Moreover, $S_1^{FB}(0) = 1/6$ and, from (2), we have $S_1^{FB}(\mu) = (1 - \mu)^2 S_1^{FB}(0)$. Thus, the ratios $CS_{\infty}^M(\mu)/S_1^{FB}(\mu)$ and $PS_{\infty}^M(\mu)/S_1^{FB}(\mu)$ are equal to $3/(16(1 - \mu))$. As illustrated in Figure 3, this is increasing in μ and larger than 1 for $\mu > 13/16 = 0.8125$. The figure also displays $(CS_{\infty}^M(\mu) + PS_{\infty}^M(\mu))/S_1^{FB}(\mu) = 3/(8(1 - \mu))$ (that is, the sum of consumer surplus per buyer and producer surplus per seller divided by first-best welfare under bilateral trade). This ratio is larger than 1 for $\mu > 5/8 = 0.625$. This shows that the matching benefits of thick monopoly markets come to bear for μ well below 1. Moreover, if one were to adopt a social surplus perspective and also take the monopoly's profit into account, the monopoly would outperform first-best bilateral trade even with mass products (i.e. for any $\mu \in [0, 1)$) since social surplus under the thick market monopoly is $3/16$, which exceeds $S_1^{FB}(0) = 1/6$ (this last point was also made in footnote 1).*

3.4 Intermediate degrees of market thickness

Up to this point we have focused on two extreme benchmarks: thin markets involving bilateral trade and perfectly thick markets involving a continuum of traders on each side of the market. In this section, we allow for intermediate degrees of market thickness by considering

markets for niche products involving n pairs of buyers and sellers. We continue to assume that each buyer independently draws their type from the distribution F_μ and each seller independently draws their type from the distribution G_μ .

3.4.1 Surplus under monopoly in finitely thick markets

We begin by providing a characterization of social surplus, consumer surplus and producer surplus under a monopolist market maker for finitely thick markets. Formally, given μ , we let $S_n^M(\mu)$ denote welfare per trader pair, $CS_n^M(\mu)$ denote the surplus per buyer and $PS_n^M(\mu)$ denote the surplus per seller under a market with a monopoly market maker and n pairs of buyers and sellers. We then have the following generalization of Proposition 3 for finitely thick markets.

Proposition 4. *Given any $n \geq 2$ and for μ sufficiently large, the ratios $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)}$, $\frac{CS_n^M(\mu)}{S_1^{FB}(\mu)}$ and $\frac{PS_n^M(\mu)}{S_1^{FB}(\mu)}$ are increasing in μ . Moreover, there exists a threshold $\hat{\mu} < 1$ such that $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} > 1$, $\frac{CS_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ and $\frac{PS_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ for all $\mu > \hat{\mu}$ and sufficiently large n .*

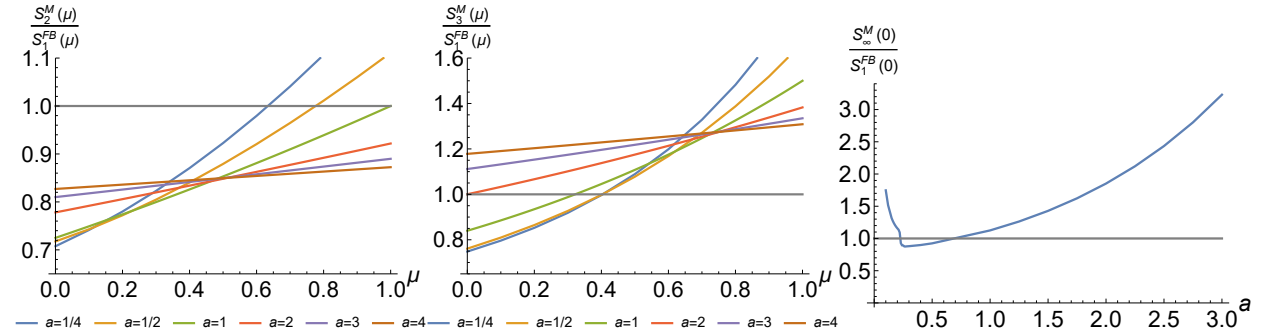


Figure 4: Each panel of this figure utilizes the beta distributions $F(v) = 1 - (1 - v)^a$ and $G(c) = c^a$. The left-hand panel illustrates the ratio $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)}$ for $n = 2$ and a variety of a values. The center panel illustrates $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)}$ for $n = 3$ and the same set of a values. The right-hand panel illustrates $\frac{S_\infty^M(\mu)}{S_1^{FB}(\mu)}$ for $\mu = 0$ and $a \in [0.1, 3]$.

Proposition 4 is illustrated in figures 4 and 5, which utilize a family of beta distributions parameterized by a single parameter $a > 0$ with $F(v) = 1 - (1 - v)^a$ and $G(c) = c^a$. These distributions admit symmetric linear virtual type functions that are such that $CS_n^M(\mu) = PS_n^M(\mu)$ for all $a > 0$ and $n \geq 2$.

Figure 4 displays a series of plots relating to social surplus. It is noteworthy how well, from a social surplus perspective, the monopoly market performs relative to ex post efficient

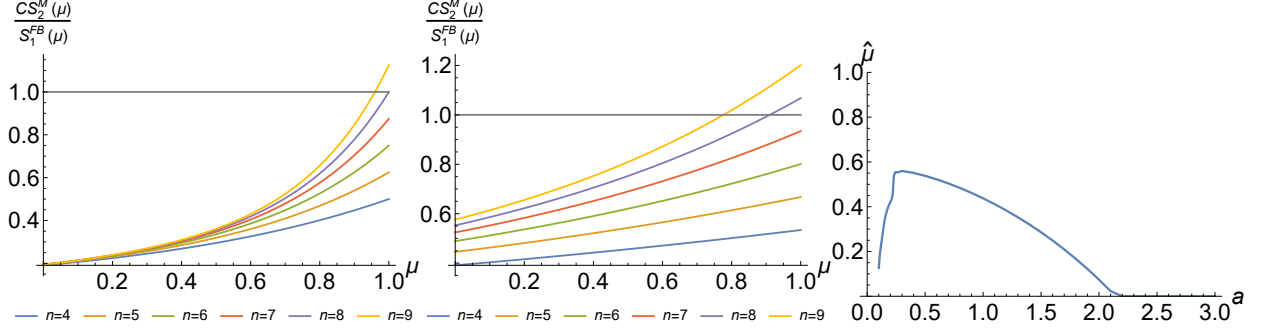


Figure 5: Each panel of this figure utilizes the beta distributions $F(v) = 1 - (1 - v)^a$ and $G(c) = c^a$ (which admit symmetric linear virtual type functions). The left-hand panel illustrates the ratio $\frac{CS_n^M(\mu)}{S_1^{FB}(\mu)} = \frac{PS_n^M(\mu)}{S_1^{FB}(\mu)}$ for $a = 1$ (i.e. uniformly distributed types) and a variety of n values. The center panel illustrates $\frac{CS_n^M(\mu)}{S_1^{FB}(\mu)} = \frac{PS_n^M(\mu)}{S_1^{FB}(\mu)}$ for $a = 3$ and a variety of n values. The right-hand panel illustrates $\hat{\mu} = \min \left\{ \mu \in [0, 1] : \frac{CS_\infty^M(\mu)}{S_1^{FB}(\mu)} = \frac{PS_\infty^M(\mu)}{S_1^{FB}(\mu)} \geq 1 \right\}$ for $a \in [0.1, 3]$.

bilateral trade even with seemingly thin markets.¹⁶ In particular, the left-hand panel shows that for $a \in \{1/4, 1/2\}$ we already have $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ for $n = 2$ and a sufficiently niche product. The center panel shows that for some distributions, $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ already holds for all $\mu \in [0, 1)$ (i.e. irrespective of product nichelessness) once $n \geq 3$. The right-hand panel of Figure 4 provides an illustration of the necessity of the threshold $\hat{\mu}$ in the statement of Proposition 4 (i.e. even with an arbitrarily large n , it is not generally the case that $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} > 1$, $\frac{CS_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ and $\frac{PS_n^M(\mu)}{S_n^{FB}(\mu)} > 1$ for all $\mu \in [0, 1)$).

Figure 5 displays a series of plots relating to consumer and producer surplus for finitely thick markets. These plots qualify that although the monopoly market maker performs favorably from a social surplus perspective for the family of distributions we consider, in very thin markets much of the benefits of market thickness are captured by the monopolist intermediary. Nevertheless, the left-hand and center panels show that there are still significant benefits accruing to participants in markets for niche products for relatively thin markets ($n \geq 8$). It is also worth noting that when considering producer and consumer surplus in thin markets, using social surplus under ex post efficient bilateral trade provides a relatively strict benchmark because it ignores the incentive issues associated with private information and runs a deficit equal to the gains from trade for our setting. The right-hand panel provides an illustration of the cutoff $\hat{\mu}$ from Proposition 4 for $a \in [0.1, 3]$. This figure illustrates that

¹⁶This observation is in a similar spirit to some of the findings of Loertscher et al. (2022), who show that storing just a single trader yields 2/3 of the total gains from market thickness (associated with storing a continuum of traders) for a specific dynamic market making setting.

for some distributions (i.e. for $a > 2.1429$), we have $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} > 1$, $\frac{CS_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ and $\frac{PS_n^M(\mu)}{S_1^{FB}(\mu)} > 1$ for all $\mu \in [0, 1]$.

3.4.2 Returns to scale of market thickness

We now turn to analyzing the returns to scale in market making for finitely thick markets. Here, a natural measure of market thickness is the proportion of the maximum increase in per trader pair surplus

$$T_n(\mu) := \frac{S_n^{FB}(\mu) - S_1^{FB}(\mu)}{S_\infty(\mu) - S_1^{FB}(\mu)}$$

that is achieved by a market with n pairs of buyers and sellers. Here, $S_\infty(\mu) - S_1^{FB}(\mu)$ captures the maximum achievable increase in welfare per trader pair as we move from a perfectly thin market with a single pair of traders to a perfectly thick market in the limit as $n \rightarrow \infty$. This measure of market thickness is analogous to that introduced in Loertscher et al. (2022). Accordingly, $T_{n+1}(\mu)/T_n(\mu)$ provides a natural measure of the returns to scale in market making.

In order to compute $T_n(\mu)$ and $T_{n+1}(\mu)/T_n(\mu)$, we start by computing expected welfare $S_n^{FB}(\mu)$ in a Walrasian market with n pairs of buyers and sellers by exploiting its recursive structure. Specifically, $S_{n+1}^{FB}(\mu)$ is given by $S_n^{FB}(\mu)$ plus the expected increase in welfare associated with adding an additional seller to a market with n buyers and n sellers plus the expected increase in welfare associated with adding an additional buyer to a market with n buyers and $n + 1$ sellers. Given a niche market with n buyers and m sellers we let $X = \{v_1, \dots, v_n, c_1, \dots, c_m\}$ denote the joint set of trader valuations and costs, $x_{(1)} \leq \dots \leq x_{(n+m)}$ denote the order statistics of this set and $h_{\mu,i,n,m}$ denote the density of the i th order statistic $x_{(i)}$, that is, the i th lowest draw. We then have the following proposition.

Proposition 5. *Welfare in a Walrasian market with $n+1$ pairs of buyers and sellers satisfies the recursion*

$$\begin{aligned} S_{n+1}^{FB}(\mu) &= S_n^{FB}(\mu) + (1 - \mu) \int_0^1 \int_0^x (x - c)g(c)h_{\mu,n,n,n}(x) dc dx \\ &\quad + (1 - \mu) \int_0^1 \int_x^1 (v - x)f(v)h_{\mu,n+1,n,n+1}(x) dv dx, \end{aligned}$$

with $S_1^{FB}(\mu) = (1 - \mu)^2 \int_0^1 \int_c^1 (v - c)f(v)g(c) dv dc$.

Figure 6 provides an illustration of the returns to scale in market making, $\frac{T_{n+1}(\mu)}{T_n(\mu)}$, for the special case of uniform distributions. Corollary B.1 in Appendix B.7 provides closed-form solutions for the expressions plotted here. The figure shows that the returns to scale in

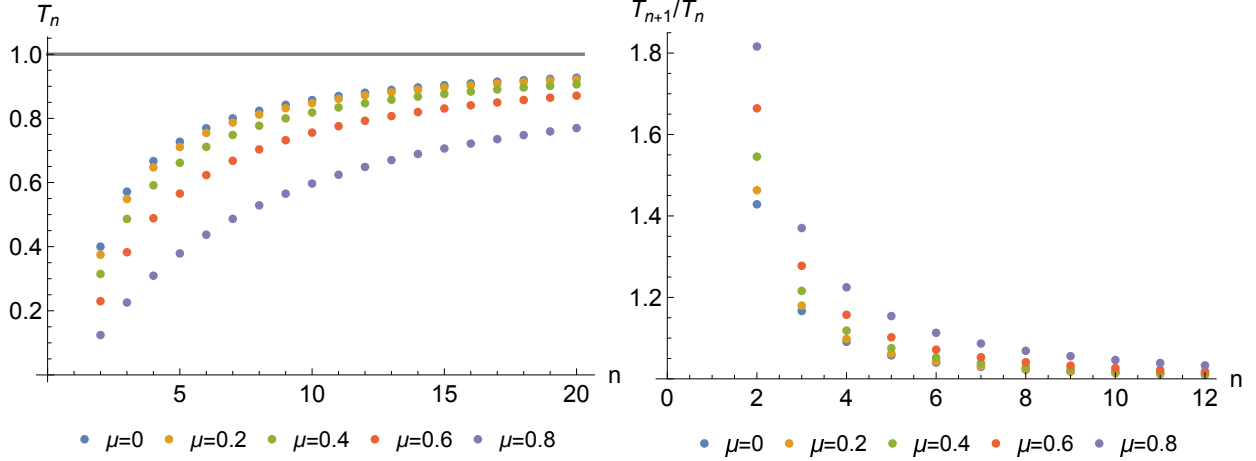


Figure 6: Illustration of our measure of market thickness T_n and returns to scale in market making T_{n+1}/T_n for uniform distributions and various values of μ and n .

market making are larger in thinner markets and for niche products. In Appendix B.5 we numerically show that these comparative statics also hold in an alternative model of niche products involving linear virtual types, where products become more niche as the elasticity of both demand and supply increase. Figure 6 highlights the importance of market thickness in markets for niche products.

4 Policy discussion

We now provide a discussion of some policy-relevant aspects of our analysis. We first consider the problem of a benevolent planner that decides whether to run a centralized, Walrasian market at some fixed cost. This analysis reveals that centralized markets are either set up for products of intermediate nicheness or for products with a degree of nicheness below some threshold. We then show that the same result applies if the thick market is set up by a profit-maximizing intermediary. Second, we consider how introducing bilateral trade as an outside option for traders impacts the market power of thick market monopolies. We show that their exertion of market power is least constrained by this outside option in markets for highly niche products, suggesting that regulatory scrutiny may be most needed for such products.

4.1 Fixed costs and centralized thick markets

A question of considerable practical relevance is the following: When are the social benefits from establishing centralized markets that reap the matching benefits of market thickness

greatest? For example, a policy maker setting up a centralized clearinghouse for healthcare services may want to know whether the social benefits of doing so are greatest for the most common healthcare services, for the most niche healthcare services or for needs that are somewhere in between. With this in mind, we now assume that establishing a centralized, thick market comes at a fixed cost of $K > 0$. A benevolent social planner or a profit-maximizing intermediary will then establish a thick market if and only if the gains from doing so outweighs the fixed cost K . Among other things, this allows us to account for the absolute (or aggregate) benefits of market thickness rather than the relative benefits that were the focus of the preceding section.

We will show that—unless K is so large that establishing a centralized market is never efficient—the social planner will either establish a centralized market for products of intermediate nicheness or establish a centralized market for all products with a degree of nicheness that is below some threshold. Consequently, a centralized market will never be established for the most niche products. This is despite the fact that (as shown in Proposition 2) the relative gains from market thickness ($S_\infty(\mu)/S_1(\mu)$) increase in μ and grow unboundedly as $\mu \rightarrow 1$. If $K > 0$, it will never be efficient to establish a centralized market for sufficiently niche products simply because the aggregate gains are too small.

Formally, we normalize the mass of buyers and sellers in the large market to 1 so that we do not have to differentiate expressions written in per trader terms from the fixed cost K and the aggregate value of gains from trade. We first assume that the decision to set up a centralized market is made for a single, known value of μ and ask when (i.e. for which values of μ) it will be efficient to set up the centralized market. Under the same assumptions, we then ask when a profit-maximizing intermediary will set up the centralized market. Finally, we discuss what happens if there is a non-degenerate distribution of μ and a single centralized market can match buyers and sellers for all values of μ . This gives rise to economies of scope that may be descriptive of platforms such as Amazon and eBay.

Because $S_1^{FB}(\mu) = (1 - \mu)^2 S_1^{FB}(0)$, $S_\infty(\mu) = (1 - \mu) S_\infty(0)$ and $s_\infty = \frac{S_\infty(0)}{S_1^{FB}(0)}$, the gain from establishing the centralized market, denoted $W(\mu)$, is

$$W(\mu) = S_\infty(\mu) - S_1^{FB}(\mu) = S_1^{FB}(0)(1 - \mu)(s_\infty - (1 - \mu)),$$

which satisfies $W'(\mu) = S_1^{FB}(0)(2(1 - \mu) - s_\infty)$, $W''(\mu) = -2S_1^{FB}(0) < 0$, $W(0) = S_\infty(0) - S_1^{FB}(0) \geq 0$ and $W(1) = 0$. That is, W is a concave function and equal to 0 at $\mu = 1$. Its derivative at $\mu = 0$ is positive if and only if $s_\infty < 2$. Hence, the maximizer of $W(\mu)$ is $\mu^* = \max\{1 - \frac{s_\infty}{2}, 0\}$. If $s_\infty < 2$, then $\mu^* > 0$ and $W(\mu^*) = \frac{1}{4} \frac{(S_\infty(0))^2}{S_1^{FB}(0)}$. Otherwise, $W(\mu^*) = W(0) = S_\infty(0) - S_1^{FB}(0)$.

If $K \geq W(\mu^*)$, then establishing the centralized market is never efficient. So assume from here onward that $K \leq W(\mu^*)$. Then we have the following result:

Proposition 6. *Assume $0 < K < W(\mu^*)$. Then there exist $\underline{\mu}$ and $\bar{\mu}$ satisfying $0 \leq \underline{\mu} < \bar{\mu} < 1$ such that the planner sets up a centralized market if and only if $\mu \in [\underline{\mu}, \bar{\mu}]$. Moreover, $\underline{\mu} = 0$ if and only if $K < W(0)$.*

The result that, for $s_\infty < 2$, the gains from centralization are largest for intermediate values of μ (and hence intermediate values of $(1 - \mu)w$) resonates with the finding in the dynamic setting of Loertscher et al. (2022) that the optimal degree of market thickness is largest when the Walrasian quantity is interior.¹⁷ In Loertscher et al. the planner generates market thickness by storing traders, and the optimal degree of market thickness is determined by a tradeoff between the opportunity cost of delayed consumption and future gains from trade. The gains from storing trades increases as the probability that a mismatched pair of traders arrives in any given period, and this probability is largest for intermediate values of the Walrasian quantity.

Figure 7 illustrates these findings for the family of distributions parameterized by $a > 0$ with $F(v) = 1 - (1 - v)^a$ and $G(c) = c^a$. In a thick Walrasian market with $\mu = 0$, the quantity traded $w = \frac{1}{2a}$ is then a decreasing function of a . The maximizer of $W(\mu)$, $\mu^*(a)$, is then decreasing in a and equal to 0 at $a \approx 1.57$. Put differently, $\mu^*(a)$ is increasing in the Walrasian quantity at $\mu = 0$.

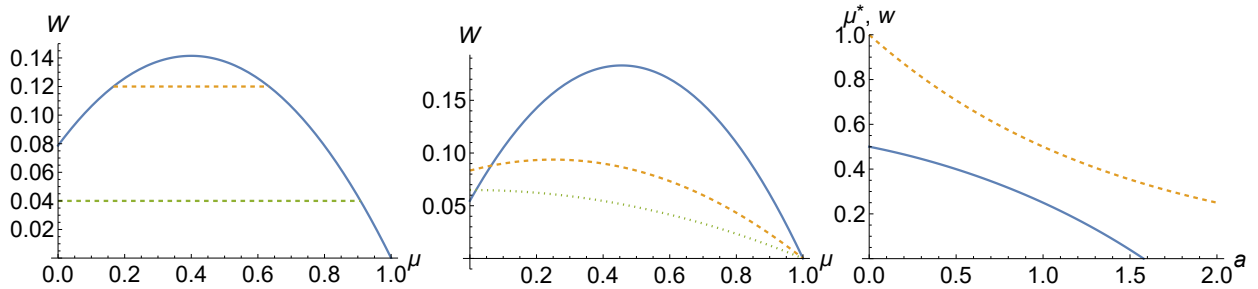


Figure 7: The left panel displays $W(\mu)$ for $a = 1/2$ and two levels of fixed costs, $K = 0.12$ and $K = 0.04$, and the associated ranges $[\underline{\mu}, \bar{\mu}]$ for which establishing the centralized market is efficient. The center panel shows $W(\mu)$ for $a = \{1/4, 1, 1.57\}$. The right panel displays w (dashed) and μ^* (solid) as a function of a .

¹⁷Loertscher et al. (2022) show that the optimal degree of market thickness, measured by a storage threshold, is largest when the parameter that represents the Walrasian quantity, denoted here by $\bar{w}$, is equal to $1/2$. Under the assumptions maintained there, $\frac{1}{s_\infty} = \bar{w} + 2(1 - \bar{w})\Delta$, where $\Delta \in (0, 1/2)$. This is larger than $1/2$ for $\bar{w} = 1/2$, which is equivalent to $s_\infty < 2$.

Profit maximization Of course, in the digital economy, trading platforms are generally established with a profit motive, rather than a social surplus motive. Assuming that the firm is a profit-maximizing intermediary for bilateral trade absent a centralized exchange, we now show that our previous insights carry over to the case where a profit-maximizing firm establishes a centralized, thick market. For this purpose, we assume for simplicity that Φ and Γ are monotone functions. For $\mu = 0$, we let $\Pi_1(0) = \mathbb{E}[(\Phi(v) - \Gamma(c))Q^1(v, c)]$ denote the intermediary's profit under bilateral trade and $\Pi_\infty(0) = m(p_B - p_S)$ denote its profit for the thick market, where p_B, p_S and m are given by (4) and (6). Given μ , the intermediary's profits in the thin and thick market are then $\Pi_1(\mu) = (1 - \mu)\Pi_1(0)$ and $\Pi_\infty(\mu) = (1 - \mu)\Pi_\infty(0)$, respectively. Consequently, its benefit $W_\pi(\mu)$ from establishing the platform is $W_\pi(\mu) = \Pi_1(0)(1 - \mu)(\pi_\infty - (1 - \mu))$, where $\pi_\infty = \frac{\Pi_\infty(0)}{\Pi_1(0)}$. Repeating our previous arguments, the maximizer of $W_\pi(\mu)$ is $\mu_\pi^* = \max\{1 - \frac{\pi_\infty}{2}, 0\}$. If $K < W(\mu^*)$, then the intermediary establishes the platform for any $\mu \in [\underline{\mu}_\pi, \bar{\mu}_\pi]$, where $0 \leq \underline{\mu}_\pi < \bar{\mu}_\pi < 1$, and it does not establish the platform otherwise. For the family of distributions parameterized by $a > 0$ with $F(v) = 1 - (1 - v)^a$ and $G(c) = c^a$, we have $\mu_\pi^* \leq \mu^*$ for all $a > 0$, with strict inequality for all a such that $\mu^* > 0$ (which corresponds to $a < 1.57$). Moreover, if $\mu_\pi^* > 0$ (which corresponds to $a < 0.45$), we have $W_\pi(\mu_\pi^*) = W(\mu^*)$ and $\frac{\pi_\infty}{s_\infty} = \left(\frac{1+a}{a}\right)^a$. The latter is an increasing function of a with $\frac{\pi_\infty}{s_\infty} \Big|_{a=1} = 1$ and $\lim_{a \rightarrow \infty} \frac{\pi_\infty}{s_\infty} = e$. Further, we have $\frac{S_1^{FB}(0)}{\Pi_1(0)} = \left(\frac{1+a}{a}\right)^{2a}$, and this last expression is greater than 1 for any $a > 0$. This implies that $\frac{W'(\mu)}{W'_\pi(\mu)} = \frac{S_1^{FB}(0)}{\Pi_1(0)} \frac{2(1-\mu)-s_\infty}{2(1-\mu)-\pi_\infty}$ which, together with $\frac{\pi_\infty}{s_\infty} > 1$, implies that $W'(\mu) > W'_\pi(\mu)$ for any $\mu \in [0, 1]$. Finally, we also have $\underline{\mu}_\pi \leq \underline{\mu}$ and $\bar{\mu}_\pi \leq \bar{\mu}$. This shows that the profit-maximizing intermediary caters more to mass products relative to a social planner.

Economies of scope To conclude this analysis, we assume that there is a non-degenerate distribution of nicheness parameters μ and that a centralized market exhibits economies of scope insofar as, once it is established, the centralized market permits running exchanges for the various products characterized by all the different values of μ . The social planner will establish the centralized market if and only if $\mathbb{E}[W(\mu)] - K > 0$ while the profit-maximizing intermediary will do so if and only if $\mathbb{E}[W_\pi(\mu)] - K > 0$. As noted, both the planner and the profit-maximizing intermediary require enough probability mass on sufficiently mass products to establish the centralized markets, with the profit-maximizer being more biased toward mass products than the planner. However, once a platform that exhibits economies of scope is established, the traders of niche products benefit from it as well. If this is the case, traders of mass products exert a positive externality on traders of niche products.

4.2 Competing bilateral exchanges constraining market power

The double coincidence of wants problem is less severe for mass products than for niche products because the probability of finding a trading partner in, say, a bilateral trade setting is larger. This suggests that bilateral trade offers a viable outside option for mass products, and less so for niche products. Since outside options reduce buyers' willingness to pay and increase sellers' reservation prices, one would expect that in mass product markets, the market power of a thick market monopoly is constrained by the outside option offered by bilateral trade. We are now going to formalize this notion by extending the model to allow for the possibility that a perfectly thick large market coexists with bilateral trade.

Without this interpretation being necessary, one can think of the large market monopoly as a digital platform and of buyers and sellers as living in otherwise disconnected, geographical markets such as different towns or countries, each of which is connected to the platform. In line with the main analysis of this paper, we think of these local markets as being characterized by bilateral trade.

To fix ideas, we assume that trade in a bilateral exchange occurs via a posted price mechanism à la Hagerty and Rogerson (1987), where the posted price is the Walrasian price p . After learning its type each agent has the option of trading via the thick market monopoly or in a bilateral exchange where trade occurs if and only if the buyer and seller are willing to trade at p . For simplicity, assume that we have a regular mechanism design problem, which is to say that F and G exhibit increasing virtual type functions. This implies that if the outside option of agents is 0, posting prices of p_B and p_S is optimal for the monopoly. Even when agents have the outside option of joining a bilateral exchange, we assume that the monopoly is restricted to posting prices.¹⁸

While there may be multiple equilibria, we focus on the equilibrium with monotone sorting in which for any $p_B > p > p_S$, all buyers with $v \in (\bar{v}, 1]$ and all sellers with $c \in [0, \underline{c})$ trade via the monopoly and all the other types join the bilateral exchanges, where $\bar{v} > p_B$ and $p_S > \underline{c}$ and $1 - F(\bar{v}) = G(\underline{c})$. Under the stipulated assumptions, the respective expected utilities of a buyer of type v and seller of type c from the bilateral exchange is

$$U_B(v) := (1 - \mu)(G(p) - G(\underline{c}))(v - p) \quad \text{and} \quad U_S(c) := (1 - \mu)(F(\bar{v}) - F(p))(p - c),$$

where $v \geq p$ and $p \geq c$. If $v \geq p_B$ and $p_S \geq c$, the payoffs of these agents from going to the monopoly are $v - p_B$ and $p_S - c$, respectively. It follows that $\bar{v}$ and $\underline{c}$ satisfy $\bar{v} - p_B = U_B(\bar{v})$

¹⁸It is an open question of what is the optimal mechanism for a monopoly when the agents have the outside option of joining bilateral exchanges.

and $p_S - \underline{c} = U_S(\underline{c})$.¹⁹ Hence, for given $\bar{v}$ and $\underline{c}$, we have $p_B = \bar{v} - U_B(\bar{v})$ and $p_S = \underline{c} + U_S(\underline{c})$, which implies that the monopoly's spread is

$$p_B - p_S = \bar{v} - \underline{c} - (U_B(\bar{v}) + U_S(\underline{c})). \quad (7)$$

Since it is never optimal to choose p_B and p_S such that $1 - F(\bar{v}) \neq G(\underline{c})$, we can replace $1 - F(\bar{v})$ and $G(\underline{c})$ by the monopoly's "quantity" $q \in [0, 1]$ (bearing in mind that if it chooses q its quantity traded will be $(1 - \mu)q$). Replacing $F(p)$ by $1 - w$ and $G(p)$ by w , the spread in (7) thus simplifies to

$$p_B - p_S = (F^{-1}(1 - q) - G^{-1}(q)) (1 - (1 - \mu)(w - q)).$$

Figure 8 illustrates the effects of changes in μ on the inverse demand and supply functions the monopoly faces when the agents have the outside option of participating in a bilateral exchange, where the demand and supply functions are normalized by $1 - \mu$. The monopoly's profit maximization problem can thus be written

$$\max_{q \in [0, 1]} (F^{-1}(1 - q) - G^{-1}(q)) (1 - (1 - \mu)(w - q))q, \quad (8)$$

which has the same logic and structure as that underlying (5). Denote the solution to this problem by $q^*(\mu)$.²⁰ Denoting the prices associated with $q^*(\mu)$ by $p_B^*(\mu)$ and $p_S^*(\mu)$, we have

$$p_B^*(\mu) = F^{-1}(1 - q^*(\mu)) - (F^{-1}(1 - q^*(\mu)) - p)(1 - \mu)(w - q^*(\mu)) \quad (9)$$

and

$$p_S^*(\mu) = G^{-1}(q^*(\mu)) + (p - G^{-1}(q^*(\mu)))(1 - \mu)(w - q^*(\mu)). \quad (10)$$

Proposition 7. *Suppose that $\pi(q) := (F^{-1}(1 - q) - G^{-1}(q))q$ is concave. Then $q^*(\mu)$ is strictly decreasing in μ with $q^*(1) = m$ and the spread $p_B^*(\mu) - p_S^*(\mu)$ is increasing in μ .*

Proposition 7 provides a concise formalization of the notion that mass products mitigate market power. As products become more niche and μ increases, the monopoly's quantity in the presence of competing bilateral exchanges decreases and its pricing becomes more aggressive. Both effects reduce the sum of consumer and producer surplus at the monopoly. If F and G are symmetric in such a way that $p_B^*(\mu) = 1 - p_S^*(\mu)$ (for example, if F and G

¹⁹Note that since $U'_B(v) < 1$ and $U'_S(c) > -1$ the single crossing condition holds.

²⁰We do not scale this profit by $1 - \mu$ to make sure the maximization problem is well defined even as $\mu \rightarrow 1$ and to isolate the effects of μ via the outside options; the objective (8) can be thought of as a normalized profit.

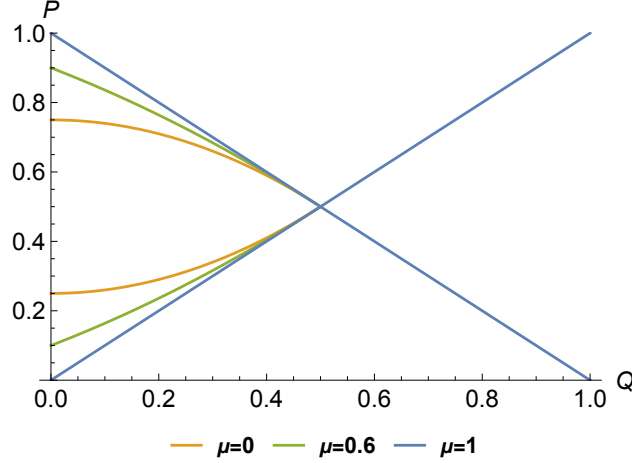


Figure 8: The inverse demand and supply functions faced by the monopoly when agents have bilateral trade as an outside option and F and G are uniform distributions.

correspond to uniform distributions), then the decreasing nature of the spread implies that $p_S^*(\mu)$ decreases in μ and $p_B^*(\mu)$ increases in μ . Thus, (normalized) consumer surplus and (normalized) producer surplus at the monopoly each decrease in μ .

The fact that $q^*(\mu) > m$ for any $\mu < 1$ is of independent interest. An increase in the monopoly's quantity decreases the probability of trade in the bilateral exchange and, consequently, the payoffs $U_B(\bar{v})$ and $U_S(\underline{c})$. Therefore, the increase in the quantity traded by the monopoly, relative to the model without bilateral exchanges, is driven by a foreclosure motive.

In Appendix B.6 we perform an analogous analysis, assuming that the second-best mechanism operates in the competing exchange and show that the key results derived here continue to hold.

A number of messages emerge from the analysis in this section. First, market power is naturally mitigated with mass products because bilateral trade offers viable outside options. Consequently, abuse of market power is less of a concern with thick market monopolies for mass products. By the same token, it is in the interest of the monopoly to drive out even small (bilateral) exchanges because these offer viable outside options and constrain its profit. Of course, in practice and in richer models, the monopoly may take additional or alternative actions to drive out small rivals. This suggests that for mass products, the main concern or focus should be on the foreclosure motive rather than on abuse of market power. Second, for niche products, small bilateral exchanges provide a less viable outside option for traders precisely because the monopoly generates more consumer and producer surplus than bilateral trade for sufficiently large μ . In other words, for niche products there is no natural mitigation of market power in the face of bilateral exchanges. Consequently, abuse

of market power is a greater concern and foreclosure of small exchanges is of less concern for niche products.²¹

5 Conclusions

In this paper, we use an independent private values model to quantify the matching benefits of thick markets relative to thin markets. We distinguish between niche and mass products and model thin markets as bilateral exchanges and thick markets as markets with a continuum of buyers and sellers. We show that whether a product is niche or mass neither affects the severity of the incentive problem in thin markets nor the severity of the harm to consumers, producers and social surplus under a thick market monopoly relative to an efficient thick market. In contrast, the consumer, producer and social surplus benefits of thick markets relative to thin markets increase as the products become increasingly niche due to the matching benefits associated with market thickness. Consequently, for products that are sufficiently niche, the benefits of thick market monopolies are first order and the harm from monopolies relative to efficient thick markets are second order. These results are driven by the fact that for niche products thick markets overcome the severe double coincidence of wants problem that arises in thin markets. By the same token we show that competing bilateral exchanges constrain the market power of thick market monopolies for mass products but not for niche products. This provides some rationale for increasing regulatory scrutiny of thick market monopolies for niche products.

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²¹ As mentioned in the introduction, the competing bilateral exchanges we analyze differ from the competing exchanges the literature has mostly focused on to date; see, for example, Gehrig (1993), Spulber (2002), Rust and Hall (2003), and Loertscher and Niedermayer (2020). Even though matching in the competing exchange in a model such as Gehrig’s is random, the competing exchange is a thick market because there is a continuum of agents, and these agents participate only if they expect positive surplus from so doing. Hence, even if matching were random, such a thick competing exchange would eliminate the double coincidence of wants problem and would thus be a severe competitive constraint for a thick market monopoly for niche products.

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Appendix

A Proofs

A.1 Proof of Lemma 1

Proof. The proof is confined to the weighted virtual value function as the proof for the weighted virtual cost function is analogous. Using $F(v) = \frac{F_\mu(v) - F_\mu(0)}{1 - F_\mu(0)}$ and $f(v) = \frac{f_\mu(v)}{1 - F_\mu(0)}$, we have $\Phi_\alpha(v) = v - \alpha \frac{1 - F(v)}{f(v)} = v - \alpha \frac{\frac{1 - F_\mu(v)}{1 - F_\mu(0)}}{\frac{f_\mu(v)}{1 - F_\mu(0)}} = v - \alpha \frac{1 - F_\mu(v)}{f_\mu(v)}$ as claimed. $\square$

A.2 Proof of Proposition 1

Proof. Recall that for any $\alpha \in [0, 1]$ we let $\bar{\Phi}_\alpha(v)$ and $\bar{\Gamma}_\alpha(c)$ denote the weighted *ironed* virtual valuation and cost functions, respectively. Due to the truncation invariance of these functions, they apply regardless of the value of the niche product parameter μ . The mechanism that maximizes the convex combination of welfare and profit with a weight of α on profit then has the allocation rule $Q^\alpha(v, c) = \mathbb{1}(\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c))$ that induces trade if and only if $\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c)$. Therefore, for any $\alpha \in [0, 1]$, the mechanism that implements the allocation rule Q^α is uniquely pinned down and does not vary with the parameter μ .²² For any $\alpha \in [0, 1]$, we thus have

$$\begin{aligned}
 S_1^\alpha(\mu) &= \mathbb{E}[(v - c)Q^\alpha(v, c)] \\
 &= \mathbb{E}[(v - c)\mathbb{1}(\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c))] \\
 &= \int_0^1 \int_0^1 (v - c)\mathbb{1}(\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c))f_\mu(v)g_\mu(c)dc dv \\
 &= (1 - \mu)^2 \int_0^1 \int_0^1 (v - c)\mathbb{1}(\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c))f(v)g(c)dc dv \\
 &= (1 - \mu)^2 \mathbb{E}[(v - c)Q^\alpha(v, c)] \\
 &= (1 - \mu)^2 S_1^\alpha(0).
 \end{aligned}$$

Using $S_1^{FB}(\mu) = (1 - \mu)^2 S_1^{FB}(0)$ then yields $\frac{S_1^\alpha(\mu)}{S_1^{FB}(\mu)} = \frac{S_1^\alpha(0)}{S_1^{FB}(0)}$, which shows that this ratio is independent of μ as required. $\square$

²²When $\alpha = 0$, we obtain a unique mechanism by taking the welfare-maximizing mechanism that otherwise maximizes the designer's profit.

A.3 Proof of Proposition 2

Proof. Using $S_\infty(\mu) = (1 - \mu)S_\infty(0)$, $S_1^{FB}(\mu) = (1 - \mu)^2 S_1^{FB}(0)$ and the definition of s_∞ yields $\frac{S_\infty(\mu)}{S_1^{FB}(\mu)} = \frac{s_\infty}{1 - \mu}$. Moreover, since $\mu < 1$ and $s_\infty > 1$, we have $\frac{S_\infty(\mu)}{S_1^{FB}(\mu)} > 1$. Taking the limit as $\mu \rightarrow 1$ yields the final statement of the proposition. $\square$

A.4 Proof Proposition 3

Proof. We begin by proving the first statement of the proposition. For the sake of notational brevity, we write CS and PS instead of CS_∞^M and PS_∞^M throughout the proof of this first statement. We have already proven the proposition for the regular case. In general (that is, without imposing regularity) the optimal quantity of the large mass product monopoly, denoted m , is such that

$$\bar{\Phi}(F^{-1}(1 - m)) = \bar{\Gamma}(G^{-1}(m)),$$

where $\bar{\Phi}(v) \equiv \bar{\Phi}_1(v)$ and $\bar{\Gamma}(c) \equiv \bar{\Gamma}_1(c)$.²³ It remains to determine the associated pricing mechanism. If $\bar{\Phi}(F^{-1}(1 - m)) = \Phi(F^{-1}(1 - m))$ and $\bar{\Gamma}(G^{-1}(m)) = \Gamma(G^{-1}(m))$, the problem is equivalent to the regular case analyzed in Section 3.3, and we are done. There are two additional generic and symmetric cases that need to be considered: (i) $\bar{\Phi}(F^{-1}(1 - m)) > \Phi(F^{-1}(1 - m))$ and $\bar{\Gamma}(G^{-1}(m)) = \Gamma(G^{-1}(m))$ and (ii) $\bar{\Phi}(F^{-1}(1 - m)) = \Phi(F^{-1}(1 - m))$ and $\bar{\Gamma}(G^{-1}(m)) < \Gamma(G^{-1}(m))$.²⁴

In case (i), sellers are paid $p_S = G^{-1}(m)$ and producer surplus is $m(1 - \mu)\mathbb{E}[p_S - c | c \leq p_S]$ (as it is in the regular case). Letting $[F^{-1}(1 - m_2), F^{-1}(1 - m_1)]$ denote the ironed range of $\bar{\Phi}$ containing $F^{-1}(1 - m)$, buyers with values above $F^{-1}(1 - m_1)$ are always served and pay $p_B^1 = \frac{m_2 - m}{m_2 - m_1} F^{-1}(1 - m_1) + \frac{m - m_1}{m_2 - m_1} F^{-1}(1 - m_2)$, while buyers with values $v \in [F^{-1}(1 - m_2), F^{-1}(1 - m_1)]$ participate in a lottery where they are served with probability $\frac{m - m_1}{m_2 - m_1}$ and pay $p_B^2 = F^{-1}(1 - m_2)$ if they obtain a unit (see Loertscher and Muir, 2022). Consumer

²³This is a straightforward generalization of the analysis in Section 3 of Loertscher and Muir (2022) to two-sided private information.

²⁴At first glance it might appear as though we need to consider a third case where $\bar{\Phi}(F^{-1}(1 - m)) > \Phi(F^{-1}(1 - m))$ and $\bar{\Gamma}(G^{-1}(m)) < \Gamma(G^{-1}(m))$. This case is knife-edge because it requires that the constant regions of the ironed virtual type functions coincide. In this case there are necessarily multiple quantities that are profit-maximizing for the monopoly and at least one such quantity coincides with a previous case. That is, without loss of generality we can restrict attention to mechanisms that involve rationing on at most one side of the market. Within the set of profit-maximizing quantities, consumer and producer surplus are maximized at the largest quantity; see display (8) in Loertscher and Muir (2022) for consumer surplus (the results for producer surplus are analogous). Our results continue to hold for any monotone selection of the optimal quantity.

surplus with mass products is (see Loertscher and Muir, 2022, p.15)

$$\overline{CS}(0) = \int_0^{m_1} F^{-1}(1-x)dx + \frac{m-m_1}{m_2-m_1} \int_{m_1}^{m_2} F^{-1}(1-x)dx - \overline{R}(m),$$

where $\overline{R}(m) = \int_0^m \overline{\Phi}(F^{-1}(1-x))dx$.

In case (ii), buyers pay $p_B = F^{-1}(1-m)$ and consumer surplus is $m(1-\mu)\mathbb{E}[v - p_B | v \geq p_B]$ (as it is in the regular case). Letting $[G^{-1}(m_1), G^{-1}(m_2)]$ denote the ironed range of $\overline{\Gamma}$ containing $G^{-1}(m)$, sellers with costs below $G^{-1}(m_1)$ are always served and are paid $p_S^1 = \frac{m_2-m}{m_2-m_1}G^{-1}(m_1) + \frac{m-m_1}{m_2-m_1}G^{-1}(m_2)$, while sellers with costs $c \in [G^{-1}(m_1), G^{-1}(m_2)]$ participate in a lottery where they are given the chance to produce with probability $\frac{m-m_1}{m_2-m_1}$ and are paid $p_S^2 = G^{-1}(m_2)$ if they do so. Accordingly, producer surplus is

$$\overline{PS}(0) = \overline{C}(m) - \int_0^{m_1} G^{-1}(x)dx - \frac{m-m_1}{m_2-m_1} \int_{m_1}^{m_2} G^{-1}(x)dx,$$

where $\overline{C}(m) = \int_0^m \overline{\Gamma}(G^{-1}(x))dx$ is the convex hull of the cost of procuring m units.

Truncation invariance ensures that cases (i) and (ii) do not vary with $\mu \in [0, 1)$. In particular, denote per buyer consumer surplus by $CS(\mu)$ and per seller producer surplus by $PS(\mu)$. Consumer and producer surplus in cases (i) and (ii) are then, respectively,

$$CS(\mu) = m(1-\mu)\mathbb{E}[v - p_B | v \geq p_B] \quad \text{and} \quad PS(\mu) = (1-\mu)\overline{PS}(0) \quad (11)$$

and

$$CS(\mu) = (1-\mu)\overline{CS}(0) \quad \text{and} \quad PS(\mu) = m(1-\mu)\mathbb{E}[p_S - c | c \leq p_S]. \quad (12)$$

Combining all of these cases we have

$$\frac{CS(\mu)}{S_1^{FB}(\mu)} = \frac{1}{1-\mu}cs \quad \text{and} \quad \frac{PS(\mu)}{S_1^{FB}(\mu)} = \frac{1}{1-\mu}ps \quad (13)$$

where

$$cs := \frac{CS(0)}{S_1^{FB}(0)} \quad \text{and} \quad ps := \frac{PS(0)}{S_1^{FB}(0)}.$$

Observe that $cs > 0$ and $ps > 0$. Thus, the ratios are increasing in μ . Letting

$$\hat{\mu} = \max\{1 - cs, 1 - ps\} \quad (14)$$

the first statement of the proposition follows.

We now prove the second statement of the proposition. Taking the limit as $\mu \rightarrow 1$ in (13) yields $\lim_{\mu \rightarrow 1} \frac{CS_{\infty}^M(\mu)}{S_{1^{FB}}^M(\mu)} = \lim_{\mu \rightarrow 1} \frac{PS_{\infty}^M(\mu)}{S_{1^{FB}}^M(\mu)} = \infty$ as required. From (11) and (12) we also see that $CS_{\infty}^M(\mu)$ and $PS_{\infty}^M(\mu)$ are directly proportional to $1 - \mu$. Moreover, our previous analysis implies that the optimal mechanism of the monopoly does not vary with μ and hence monopoly profits are also directly proportional to $1 - \mu$. Combining this with the expressions for $CS_{\infty}(\mu)$, $PS_{\infty}(\mu)$ and $S_{\infty}(\mu)$ provided in Section 3.3 shows that the ratios $\frac{CS_{\infty}^M(\mu)}{CS_{\infty}(\mu)}$, $\frac{PS_{\infty}^M(\mu)}{PS_{\infty}(\mu)}$ and $\frac{S_{\infty}^M(\mu)}{S_{\infty}(\mu)}$ are finite and independent of μ for $\mu \in [0, 1)$, which concludes the proof. $\square$

A.5 Proof of Proposition 4

Proof. We let $V_n = \{v_1, \dots, v_n\}$ and $C_n = \{c_1, \dots, c_n\}$ denote the sets of buyer valuations and seller costs. We denote by $v_{[1]} \geq \dots \geq v_{[n]}$ the order statistics of the set V_n , so that $v_{[i]}$ is the i th highest buyer valuation. Similarly, we denote by $c_{[1]} \leq \dots \leq c_{[n]}$ the order statistics of the set C_n , so that $c_{(i)}$ is the i th lowest seller cost. For $i \in \{1, \dots, n\}$ we let $F_{\mu, i, n}$ denote the joint distribution of the order statistics $(v_{[1]}, \dots, v_{[i]})$ and $G_{\mu, i, n}$ denote the joint distribution of the order statistics $(c_{(1)}, \dots, c_{(i)})$. The expressions for these order statistic distributions are standard (see, for example, David, 1970). We then have

$$\begin{aligned}
S_n^M(\mu) &= \int_{\Phi^{-1}(0)}^1 \int_0^{\Gamma^{-1}(\Phi(v_{[1]}))} (\Phi(v_{[1]}) - \Gamma(c_{(1)})) dG_{\mu, 1, n}(c_{(1)}) dF_{\mu, 1, n}(v_{[1]}) \\
&+ \sum_{i=2}^n \int_{\substack{v_{[1]} \in [\Phi^{-1}(0), 1], \\ c_{(1)} \in [0, \Gamma^{-1}(\Phi(v_{[1]})]), \\ v_{[2]} \in [\Phi^{-1}(\Gamma(c_{(1)})), v_{[1]}], \\ c_{(2)} \in [c_{(1)}, \Gamma^{-1}(\Phi(v_{[2]})]), \\ \vdots \\ v_{[i]} \in [\Phi^{-1}(\Gamma(c_{(i-1)})), v_{[i-1]}], \\ c_{(i)} \in [c_{(i-1)}, \Gamma^{-1}(\Phi(v_{[i]})])}} (\Phi(v_{[i]}) - \Gamma(c_{(i)})) dF_{\mu, i, n}(v_{[1]}, \dots, v_{[i]}) dG_{\mu, 1, n}(c_{(1)}, \dots, c_{(i)}).
\end{aligned}$$

For each order statistic included in the integrals in the previous expression, we pick up a factor of $(1 - \mu)$. So the first integral (which just depends on the order statistics $v_{[1]}$ and $c_{(1)}$) is of order $(1 - \mu)^2$, the second integral is of order $(1 - \mu)^4$ and so on. In particular, we

have

$$\begin{aligned}
S_n^M(\mu) &= n^2(1-\mu)^2 \int_{\Phi^{-1}(0)}^1 \int_0^{\Gamma^{-1}(\Phi(v_{[1]}))} (\Phi(v_{[1]}) - \Gamma(c_{(1)})) f(v_{[1]}) [(1-\mu)F(v_{[1]}) + \mu]^{n-1} \\
&\quad \times g(c_{(1)}) [1 - (1-\mu)G(c_{(1)})]^{n-1} dc_{(1)} dv_{[1]} + O((1-\mu)^4) \\
&= n^2\mu^{n-1}(1-\mu)^2 \int_{\Phi^{-1}(0)}^1 \int_0^{\Gamma^{-1}(\Phi(v_{[1]}))} (\Phi(v_{[1]}) - \Gamma(c_{(1)})) f(v_{[1]}) g(c_{(1)}) dc_{(1)} dv_{[1]} + O((1-\mu)^3)
\end{aligned}$$

and the ratio $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} = \frac{S_n^M(\mu)}{(1-\mu)^2 S_1^{FB}(0)}$ can be written

$$\frac{S_n^M(\mu)}{S_1^{FB}(\mu)} = \frac{n^2\mu^{n-1}}{S_1^{FB}(0)} \int_{\Phi^{-1}(0)}^1 \int_0^{\Gamma^{-1}(\Phi(v_{[1]}))} (\Phi(v_{[1]}) - \Gamma(c_{(1)})) f(v_{[1]}) g(c_{(1)}) dc_{(1)} dv_{[1]} + O(1-\mu).$$

From this last expression, we see that in the limit as $\mu \rightarrow 1$, the ratio $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)}$ can be approximated by an expression proportional to μ^{n-1} and, consequently, $\frac{S_n^M(\mu)}{S_1^{FB}(\mu)}$ is increasing in μ for sufficiently large μ . This proves the first statement of Proposition 4. The second statement of the proposition follows immediately from Proposition 3. $\square$

A.6 Proof of Proposition 5

Proof. Suppose that we have a market with n buyers with valuations $v_1, \dots, v_n$ and m sellers with costs $c_1, \dots, c_m$. Observing that an ex post efficient allocation gives the m objects to the m agents with the largest types in the set $X = \{v_1, \dots, v_n, c_1, \dots, c_m\}$, it follows that a price is market-clearing if and only if it is between the m th and $m+1$ st highest elements in X , which is equivalent being between the n th lowest and $n+1$ st lowest elements of X . Thus, the lowest Walrasian price given X is $\underline{p}(X) := x_{(n)}$ and the highest Walrasian price is $\bar{p}(X) := x_{(n+1)}$. Notice that $\bar{p}(X)$ is either a buyer's value or a seller's cost, and likewise for $\underline{p}(X)$. We then have the following well-known observation that $\bar{p}(X)$ is the social marginal product of a seller with cost $c < \bar{p}(X)$ who arrives to a market characterized by X , since either the arriving seller replaces another seller with a higher cost (which happens if $\bar{p}(X)$ is a cost) or it expands the quantity traded by one unit (which happens if $\bar{p}(X)$ is a value). If the arriving seller's cost is above $\bar{p}(X)$, its social marginal product is 0. Likewise, $\underline{p}(X)$ is the social marginal cost of a buyer with value $v > \underline{p}(X)$ who arrives to a market characterized by X , since either the arriving buyer replaces another buyer with a lower value (which happens if $\underline{p}(X)$ is a value) or it expands the quantity traded by one unit (which happens if $\underline{p}(X)$ is a cost). If the arriving buyer's value is below $\underline{p}(X)$, its social marginal product is 0. Thus, with an additional buyer with value v social surplus increases by $\max\{v - \underline{p}(X), 0\}$ and with

an additional seller with cost c , social surplus increases by $\max\{\bar{p}(X) - c, 0\}$. These last observations follow from the fact that with single-unit demands and supplies, $\underline{p}(X)$ are the trading buyers' and $\bar{p}(X)$ are the trading sellers' VCG payments, which represent the social marginal cost and products of these agents.

If an additional seller with cost c is added to a market with n buyers and n sellers, its expected welfare contribution is thus

$$\mathbb{E}[(x_{(n)} - c)\mathbb{1}(c \leq x_{(n)})] = \int_0^1 \int_0^x (x - c)g_\mu(c)h_{\mu,n,n,n}(x)dc dx.$$

Similarly, if an additional buyer with valuation v is then added to a market with n buyers and $n + 1$ sellers, its expected welfare contribution is thus

$$\mathbb{E}[(v - x_{(n+1)})\mathbb{1}(v \geq x_{(n+1)})] = \int_0^1 \int_x^1 (v - x)f_\mu(v)h_{\mu,n+1,n,n+1}(x)dv dx.$$

Putting all of this together yields the statement in the proposition. $\square$

A.7 Proof of Proposition 6

Proof. The convexity from the set $(\underline{\mu}, \bar{\mu})$ over which it is optimal to set up a centralized market follows from the concavity of W and $\bar{\mu} < 1$ from the fact that $W(1) = 0$. We are left to show that (i) $\underline{\mu} = 0$ if and only if $K < W(0)$ and that (ii) $\bar{\mu} > 0$. If $K < W(0)$, then it pays off to establish the centralized market at $\mu = 0$ and if $K \geq W(0)$, then $\mu^* > 0$ has to be the case because $K < W(\mu^*)$ by assumption. Thus, by continuity there exists a $\underline{\mu} > 0$ such that $W(\underline{\mu}) = K$. This proves (i). By concavity of H (which implies continuity), there exists an $\bar{\mu} > \underline{\mu}$ such that $W(\bar{\mu}) = \min\{K, W(\underline{\mu})\}$, which proves (ii). $\square$

A.8 Proof of Proposition 7

Proof. We start by noting that the profit function in (8) can be rewritten as

$$\Pi(q) := \pi(q)(1 - (1 - \mu)(w - q)).$$

Note that for $\mu \in [0, 1]$ we must have $q^*(\mu) \in [0, w]$ by the definition of w , the monotonicity of the supply and demand functions and the fact that the monopoly always optimally charges a positive bid-ask spread. This also implies that for any $q \in [0, w]$, we have $\pi(q) > 0$.

Differentiating the profit function with respect to q yields

$$\Pi'(q) = \pi'(q)(1 - (1 - \mu)(w - q)) + \pi(q)(1 - \mu).$$

By construction, $\pi'(m) = 0$ and $\pi(m) > 0$. Thus, for $\mu \in [0, 1)$, we have $\Pi'(m) > 0$. Suppose that $\mu \in [0, 1)$ and that $q^* < w$ satisfies the first-order condition $\Pi'(q^*) = 0$. This implies that $(1 - \mu)\pi(q^*) > 0$ and $1 - (1 - \mu)(w - q) > 0$ and, consequently, $\pi'(q^*) < 0$. Taking the second derivative of the profit function with respect to q , we have

$$\Pi''(q) = \pi''(q)(1 - (1 - \mu)(w - q)) + 2\pi'(q)(1 - \mu).$$

Then if $\mu \in [0, 1)$ and q^* satisfies $\Pi'(q^*) = 0$, we have $\pi'(q^*) < 0$ (by our previous observation) and $\pi''(q^*) \leq 0$ (since, by assumption, π is a concave function). Combining these facts shows that if $\Pi'(q) = 0$ holds, then we necessarily have $\Pi''(q) < 0$. Thus, Π is a quasiconcave function and the second-order condition for a maximum is satisfied whenever the first-order condition holds.

Now that we have established the first-order condition that characterizes q^* , we determine how q^* varies with μ . Differentiating the first-order condition

$$\Pi'(q) = \pi'(q^*(\mu))(1 - (1 - \mu)(w - q^*(\mu))) + (1 - \mu)\pi(q^*(\mu)) = 0$$

with respect to μ and dropping the argument of $q^*(\mu)$ for the sake of brevity yields

$$\pi'(q^*) \left(w - q^* + (1 - \mu) \frac{dq^*}{d\mu} \right) + \pi''(q^*)(1 - (1 - \mu)(w - q^*)) \frac{dq^*}{d\mu} = 0$$

Rearranging this expression, we have

$$[\pi''(q^*)(1 - (1 - \mu)(w - q^*)) + 2(1 - \mu)\pi'(q^*)] \frac{dq^*}{d\mu} = -(\pi'(q^*)(w - q^*) - \pi(q^*)),$$

and hence

$$\frac{dq^*}{d\mu} = -\frac{\pi'(q^*)(w - q^*) - \pi(q^*)}{\pi''(q^*)(1 - (1 - \mu)(w - q^*)) + 2(1 - \mu)\pi'(q^*)}.$$

Suppose $\mu \in [0, 1)$. Then since π is concave (which implies that $\pi''(q^*) \leq 0$), $\pi'(q^*) < 0$ and $q^* < w$ we have $\pi''(q^*)(1 - (1 - \mu)(w - q^*)) + 2(1 - \mu)\pi'(q^*) < 0$. Moreover, since $\pi'(q^*) < 0$, we also have $\pi'(q^*)(w - q^*) - \pi(q^*) < 0$. Putting all of this together yields $\frac{dq^*}{d\mu} < 0$ and $q^*(\mu)$ is strictly decreasing in μ as required.

We are left to establish that the spread $p_B^*(\mu) - p_S^*(\mu)$ is increasing in μ . To see this, let

$$\pi^{BT}(q, \mu) := (F^{-1}(1 - q) - G^{-1}(q)) (1 - (1 - \mu)(w - q))q$$

be the profit function in the presence of bilateral exchanges. A revealed preference argument shows that $\pi^{BT}(q^*(\mu), \mu)$ is increasing in μ . To see this, consider $\mu_0 < \mu_1$. Then because the monopoly could keep the quantity at $q^*(\mu_0)$ when the parameter is $\mu_1 > \mu_0$, we have

$$\begin{aligned} \pi^{BT}(q^*(\mu_1), \mu_1) &> \pi^{BT}(q^*(\mu_0), \mu_1) \\ &= (F^{-1}(1 - q^*(\mu_0)) - G^{-1}(q^*(\mu_0))) (1 - (1 - \mu_1)(w - q^*(\mu_0)))q^*(\mu_0) \\ &> (F^{-1}(1 - q^*(\mu_0)) - G^{-1}(q^*(\mu_0))) (1 - (1 - \mu_0)(w - q^*(\mu_0)))q^*(\mu_0) \\ &= \pi^{BT}(q^*(\mu_0), \mu_0). \end{aligned}$$

Hence, $\pi^{BT}(q^*(\mu), \mu)$ is increasing in μ . By definition,

$$\pi^{BT}(q^*(\mu), \mu) = (p_B^*(\mu) - p_S^*(\mu))q^*(\mu).$$

Since $q^*(\mu)$ decreases in μ , $p_B^*(\mu) - p_S^*(\mu)$ must increase in μ for $\pi^{BT}(q^*(\mu), \mu)$ to be increasing in μ . $\square$

B Extensions and robustness

The setup presented in Section 2 was constructed so that the virtual valuation and cost functions were invariant to the niche product parameter μ . This feature together with the symmetry of the setup simplified the technical aspects of the analysis whilst providing a setting suitable for elucidating the key messages of this paper. However, the main insights and results of this paper extend beyond this setting as we now show.

B.1 Asymmetric masses of buyers and sellers

The first assumption that we can easily relax is the assumption that there is an equal mass of buyers and sellers on each side of the market in the thick Walrasian market setting analyzed in Section 3.2. In particular, without loss of generality, we can always normalize the mass of agents on one side of the market, say buyers, to 1. Let $\sigma > 0$ denote the mass of sellers. Fixing $\sigma > 0$, we then immediately recover the results of Proposition 1, Proposition 2 and Proposition 3. Graphically, we accommodate this extension by appropriately rescaling the inverse supply curves illustrated in Figure 2. That is, for each value of $\mu \in [0, 1)$ we replace

the inverse supply curve $P^S\left(\frac{\cdot}{1-\mu}\right)$ with the rescaled inverse supply curve $P^S\left(\frac{\cdot}{\sigma(1-\mu)}\right)$. This leaves us with a Walrasian price $p(\sigma)$ and quantity $w(\sigma)$ that do not vary with μ and, consequently, all of our results carry over to this setting.

B.2 Asymmetric niche product parameters

The next assumption that we can readily relax is the assumption that a symmetric niche product parameter μ applies to both sides of the market. Specifically, we can allow for a niche product parameter of μ_B that applies to the buyers' side of the market and a parameter of μ_S that applies to the sellers' side of the market. In this asymmetric setting, the Walrasian quantity traded and price now vary with both μ_B and μ_S . However, the virtual type functions are unaffected. We recover results analogous to those stated in Proposition 1, Proposition 2 and Proposition 3 if we take the limit as $\mu_B \rightarrow 1$ and $\mu_S \rightarrow 1$ in a manner such that the ratio $(1 - \mu_B)/(1 - \mu_S)$ is fixed.

B.3 Varying overlapping support of distributions

Our setup in Section 2 was constructed so that for any value of $\mu \in [0, 1]$, the region of overlapping support for the distributions F_μ and G_μ was always the interval $[0, 1]$. In this section we show that the spirit of our analysis in Section 3 extends to an alternative setting where the region of overlapping support varies with μ .

We again let μ denote the niche market parameter and construct the setup so that a product becomes increasingly niche as μ increases. However, in this alternative setup we take $\mu \in (-1, 0]$. When $\mu = 0$ we now assume that buyer values $v \in [0, 1]$ are drawn from a distribution F with density f that has full support on $[0, 1]$ and seller costs $c \in [0, 1]$ are drawn from a distribution G with density g that has full support on $[0, 1]$. In the market for a niche product with parameter $\mu \in (-1, 0)$ we assume that buyer values $v \in [a_\mu, 1]$ with $a_\mu > 0$ are drawn from a distribution F_μ with density f_μ that has full support on $[a_\mu, 1]$ and seller costs $c \in [0, b_\mu]$ with $b_\mu < 1$ are drawn from a distribution G_μ with density g_μ that has full support on $[0, b_\mu]$. We further assume that, for all $\mu \in (-1, 0)$, a_μ and b_μ are such that $F(a_\mu) = -\mu = 1 - G(b_\mu)$ and that, for all $v \in [a_\mu, 1]$ and $c \in [0, b_\mu]$,

$$F_\mu(v) = \frac{F(v) + \mu}{1 + \mu} \quad \text{and} \quad G_\mu(c) = \frac{G(c)}{1 + \mu}.$$

Notice that this implies that a_μ is decreasing in μ and b_μ is increasing in μ . We let $\underline{\mu}$ denote the largest value of $\mu \in (-1, 0]$ such that $a_\mu \geq b_\mu$.

In this setup, in a market for a product with niche parameter $\mu \in (-1, 0)$, buyers values

are truncated from below so that buyer values are distributed on the interval $[a_\mu, 1]$ and seller costs are distributed on the interval $[0, b_\mu]$, where a_μ is decreasing in μ and b_μ is increasing in μ . This implies that increasing μ results in less favourable distributions in the sense that, for all $\mu, \mu' \in (-1, 1)$ with $\mu' > \mu$, $v \in [0, 1]$ and $c \in [0, 1]$, $F_{\mu'}(v) \geq F_\mu(v)$ and $G_\mu(c) \geq G_{\mu'}(c)$. We let Φ and Γ respectively denote the virtual valuation and virtual cost function that correspond to the distributions F and G . The truncation invariance of these functions imply that the same virtual valuation and virtual cost functions are valid for any $\mu \in (-1, 0)$, on the appropriately restricted domain. Observe that for $\mu \leq \underline{\mu}$, the two supports do not overlap. This implies that full trade is efficient and the first-best outcome is possible in the bilateral trade problem. Consequently, for $\mu \leq \underline{\mu}$, the Walrasian quantity traded is 1 and

$$\frac{S_1^{SB}(\mu)}{S_1^{FB}(\mu)} = \frac{S_\infty(\mu)}{S_1^{FB}(\mu)} = 1.$$

Thus, there are no benefits from thicker markets.

For all $\mu > \underline{\mu}$ we have $a_\mu < b_\mu$. Since the distribution supports overlap in this case, full trade is not efficient and the impossibility result of Myerson and Satterthwaite (1983) holds. Recall that for a given $\alpha \in [0, 1]$, we let $\bar{\Phi}_\alpha(v)$ and $\bar{\Gamma}_\alpha(c)$ denote, respectively, the weighted ironed virtual valuation and cost functions and $Q^\alpha(v, c) = \mathbb{1}(\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c))$ denote the allocation rule that induces trade if and only if $\bar{\Phi}_\alpha(v) \geq \bar{\Gamma}_\alpha(c)$. Recall also that the second-best mechanism maximizes equally weighted ex ante expected social surplus subject to incentive compatibility, individual rationality and no-deficit constraints. This mechanism is therefore characterized by the unique value of α such that the allocation rule $Q^\alpha(v, c)$ generates zero budget surplus. That is, the second-best mechanism is characterized by the unique value α_μ^* satisfying

$$\mathbb{E}[(\Phi(v) - \Gamma(c))Q^\alpha(v, c)] = 0.$$

In the setup from Section 2, the second-best mechanism (and hence α_μ^*) was invariant to μ . Here, as μ increases in value beyond $\underline{\mu}$, the Walrasian quantity traded $w(\mu)$ will decrease in μ and α_μ^* will become strictly positive and monotonically increase in μ . Formally, we have the following result.

Proposition B.1. *The parameter α_μ^* that characterizes the second-best mechanism increases in $\mu \in (-1, 0]$ with $\alpha_\mu^* = 0$ for all $\mu \leq \underline{\mu}$. Moreover, there exists $\bar{\mu} < 0$ such that $\alpha_\mu^* = \alpha_0^*$ for all $\mu \geq \bar{\mu}$.*

Proof. Let $\mu, \mu' \in (-1, 0]$ with $\mu' > \mu$. Let $R_1^\alpha(\mu)$ denote revenue under the allocation rule

Q^α . Then by construction we have

$$R_1^{\alpha_\mu^*}(\mu) = 0$$

and

$$R_1^{\alpha_\mu^*}(\mu') = \int_0^1 \int_0^1 (\Phi(v) - \Gamma(c)) \mathbb{1}(\Phi_{\alpha_\mu^*}(v) \geq \Gamma_{\alpha_\mu^*}(c)) dF_{\mu'}(v) dG_{\mu'}(c).$$

Since the integrand $(\Phi(v) - \Gamma(c)) \mathbb{1}(\Phi_{\alpha_\mu^*}(v) \geq \Gamma_{\alpha_\mu^*}(c))$ is increasing in v and decreasing in c , the distribution F_μ first-order stochastically dominates the distribution $F_{\mu'}$ and the distribution $G_{\mu'}$ first-order stochastically dominates the distribution G_μ , we have

$$R_1^{\alpha_\mu^*}(\mu') \leq R_1^{\alpha_\mu^*}(\mu) = 0.$$

Hence, $\alpha_{\mu'}^* \geq \alpha_\mu^*$ as required. Since full trade is optimal for any $\mu \leq \underline{\mu}$, we have $\alpha_\mu^* = 0$ for all $\mu \leq \underline{\mu}$. Moreover, by the impossibility result of Myerson and Satterthwaite (1983), $\alpha_0^* > 0$. This implies that when $\mu = 0$, there exists cutoff types $\underline{v}^{SB} > 0$ and $\bar{c}^{SB} < 1$ such that any buyer with $v \leq \underline{v}^{SB}$ and any seller with $c \geq \bar{c}^{SB}$ never trade under the second-best mechanism. Thus, setting $\bar{v} = \min\{\mu \in (-1, 0) : a_\mu \leq \underline{v}^{SB}, b_\mu \geq \bar{c}^{SB}\}$ we have $\alpha_\mu^* = \alpha_0^*$ for all $\mu \geq \bar{\mu}$ as required. $\square$

Proposition B.1 implies that $\frac{S_1^{SB}(\mu)}{S_1^{FB}(\mu)}$ decreases in μ . Since $w(\mu)$ is less than 1 and decreasing μ , it also follows that $\frac{S_\infty(\mu)}{S_1^{FB}(\mu)}$ increases in μ . Interestingly, α_μ^* will cease increasing in μ before μ reaches 0, that is, before $a_\mu = 0$ and $b_\mu = 1$. Indeed, letting α^* denote the second-best mechanism parameter that corresponds to the case $\mu = 0$, we have

$$\alpha_\mu^* = \alpha^*$$

for all $\mu \geq \max\{\mu_0, \mu_1\}$, where μ_0 and μ_1 are such that $\Phi_{\alpha^*}(a_{\mu_0}) = 0$ and $\Gamma_{\alpha^*}(b_{\mu_1}) = 1$. For example, when F and G are uniform, we have $\mu_0 = \mu_1 = 1/4$ with $a_{\mu_0} = 1/4$ and $b_{\mu_1} = 3/4$.

To summarize, this extension merely increases the complexity of the model without altering the main messages that emerge from the analysis in Section 3.

B.4 Exponential distributions

An alternative way we can model niche products is to consider distributions with unbounded support. We can then create increasingly niche products by shifting probability mass towards the tails of the distributions in an appropriate fashion. For example, suppose we take $v \in$

$[0, \infty)$ with $F(v) = 1 - e^{-v}$ and $c \in [\log(a), \infty)$ with $G(c) = 1 - ae^{-c}$, where $a > 0$ is a constant. This specification implies that in a perfectly thick market with a continuum of traders the Walrasian price and quantity traded are given by $p(a) = \ln(1 + a)$ and $w(a) = 1/(1 + a)$, respectively. Increasing the parameter a pushes more probability mass towards the right tail of the sellers' cost distribution and leads to a decrease in the Walrasian quantity. Thus, the parameter a plays a similar role to the parameter μ in the model from Section 2. Expressed in terms of a , we have

$$S_1^{FB}(a) = \begin{cases} \frac{1}{2}(a - 2\ln(a)), & 0 < a \leq 1 \\ \frac{1}{2a}, & a > 1 \end{cases}$$

and

$$S_\infty(a) = \ln(1 + a) - \ln(a).$$

This implies

$$\frac{S_\infty(a)}{S_1^{FB}(a)} = \begin{cases} \frac{2(\ln(1+a) - \ln(a))}{(a - 2\ln(a))}, & 0 < a \leq 1 \\ 2a(\ln(1 + a) - \ln(a)), & a > 1. \end{cases}$$

Since $a = 0$ corresponds to having $w = 1$ we have

$$\lim_{a \rightarrow 0} \frac{S_\infty(a)}{S_1^{FB}(a)} = 1.$$

Moreover, we also have

$$\lim_{a \rightarrow \infty} \frac{S_\infty(a)}{S_1^{FB}(a)} = 2.$$

Solving for the second-best mechanism, in this setting we have

$$\Phi_\alpha(v) = v - \alpha \quad \text{and} \quad \Gamma_\alpha(c, a) = \alpha \left(\frac{e^c}{a} - 1 \right) + c.$$

The value α_a^* that characterizes the second-best mechanism (see the previous subsection for a more detailed explanation of this) then satisfies

$$\frac{2}{3}e^{-\alpha}(\alpha + e^\alpha(\alpha - 2)\alpha(\text{Chi}(\alpha) - \text{Shi}(\alpha)) - 1) = 0,$$

where the functions $\text{Chi}(\cdot)$ and $\text{Shi}(\cdot)$ denote the hyperbolic cosine integral and the hyperbolic sine integral, respectively. Numerical calculations then show that the ratio $\frac{S_1^{SB}(a)}{S_1^{FB}(a)}$ is decreasing in a and is bounded from below by 0.854367.

B.5 Linear virtual types

An alternative modelling approach is to consider settings in which products become more niche as the elasticity of supply and demand increases. To fix ideas we consider a setting with linear virtual type functions in which a single parameter $a > 0$ controls the elasticity of both demand and supply, as well as the Walrasian quantity. Specifically, we assume that buyers draw their values $v \in [0, 1]$ from the distribution $F_a(v) = 1 - (1 - v)^a$ and sellers draw their costs $c \in [0, 1]$ from the distribution $G_a(c) = c^a$. This yields a Walrasian quantity of $w(a) = 2^{-a}$ and price of $p = 1/2$.

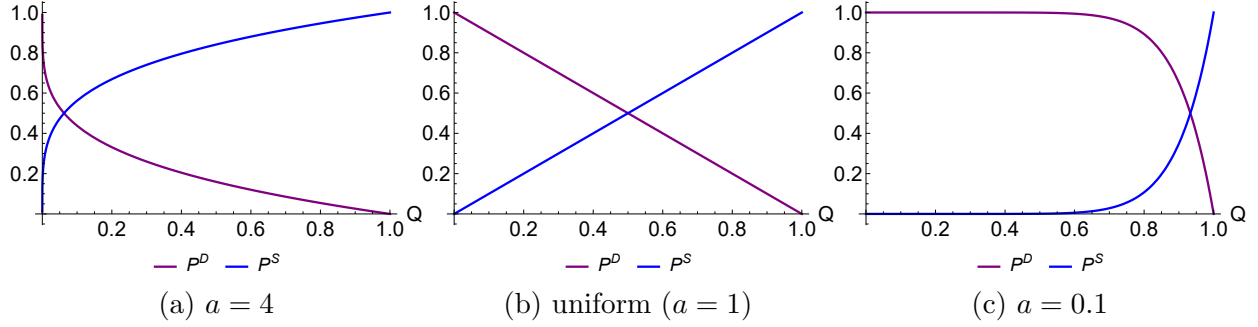


Figure 9: Demand and supply for the specification with linear virtual types.

Thus, this model exhibits similar behavior with respect to the parameter a as the model in the main body does with respect to the parameter μ . Here, as a increases and a product becomes more niche, the demand and supply curves become increasingly elastic and the Walrasian quantity $w(a)$ decreases. Per trader welfare in the thick market limit is $S_\infty(a) = \int_0^{w(a)} (F^{-1}(1 - q) - G^{-1}(q)) dq$. Figure 9 displays demand and supply for three different values of a . As a becomes small (large), the Walrasian quantity goes to $1(0)$.

In this setup, the weighted virtual type functions are given by

$$\Phi_{a,\alpha}(v) = v - \alpha \frac{1 - v}{a} \quad \text{and} \quad \Gamma_{a,\alpha}(c) = c + \alpha \frac{c}{a}.$$

Although the weighted virtual type functions and hence the ratio $\frac{S_1^{SB}(a)}{S_1^{FB}(a)}$ now vary with the parameter a , we can still bound the relative cost of incentives.

Proposition B.2. *The ratio $S_1^{SB}(a)/S_1^{FB}(a)$ is decreasing in a for $a \geq 0$ and is bounded below by $2/e \approx 0.73$. In contrast, $\lim_{a \rightarrow \infty} \frac{S_\infty(a)}{S_1^{FB}(a)} = \infty$.*

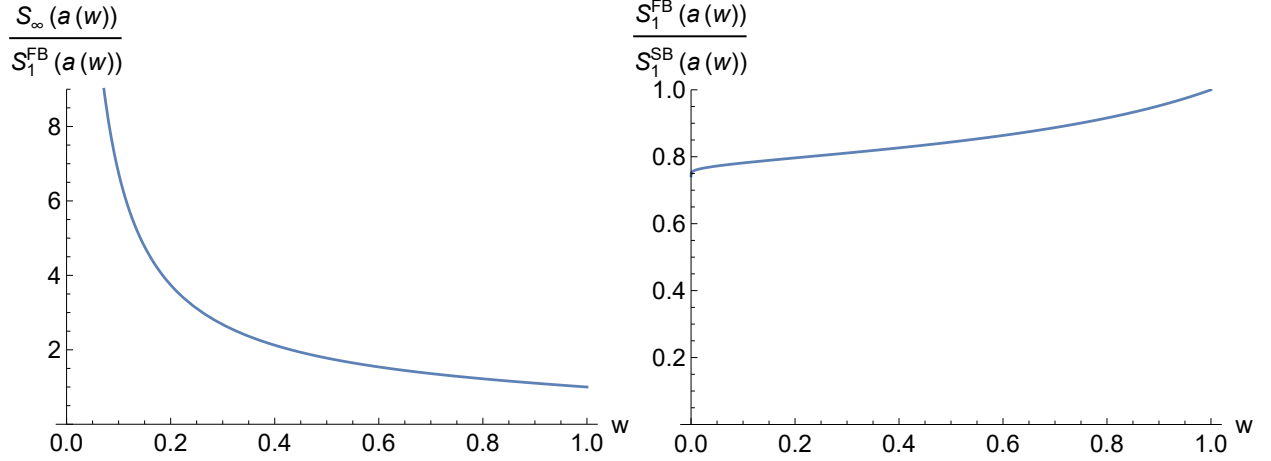


Figure 10: The matching benefits $S_\infty(a)/S_1^{FB}(a)$ of thick markets are unbounded as the Walrasian quantity goes to 0 (panel a). In contrast, the incentive benefits $S_1^{FB}(a)/S_1^{SB}(a)$ are bounded (panel b). In both panels we use $a = -\ln(w)/\ln(2)$.

Proof. Routine calculations yield

$$\frac{S_1^{SB}(a)}{S_1^{FB}(a)} = \frac{4^{-a} \left(\frac{a+1}{2a+1}\right)^{-2a} \Gamma(2a+2)}{(a+1)\Gamma(2a+1)} \quad \text{and} \quad \frac{S_\infty(a)}{S_1^{FB}(a)} = \frac{2^{-a}\Gamma(2+2a)}{a\Gamma(a)\Gamma(2+a)},$$

where (in a slight abuse of notation) here Γ denotes the gamma function. This is decreasing in a for $a \geq 0$ and taking the limit as $a \rightarrow \infty$ yields $2/e$. $\square$

With regard to the thick Walrasian market, Proposition 2 still holds as stated for this setup if we replace the limit $\mu \rightarrow 1$ with the limit $a \rightarrow \infty$. While the invariance result that holds for the parameterization with truncation of Proposition 1 does not extend to the setting with linear virtual types, it remains to be the case that the variation in $S_1^{SB}(a)/S_1^{FB}(a)$ as one varies a is bounded—it goes from 0.73 as $a \rightarrow \infty$ to 1 as $a \rightarrow 0$ —and dwarfed by the variation in S_∞/S_1^{FB} .

Consider now the thick market monopoly. Since the virtual type functions are linear (and hence monotone), the optimal trading mechanism consists of a posted price $p_B(a)$ for buyers and $p_S(a)$ for sellers. Since both the demand and supply schedules become increasingly elastic as a increases, the spread $p_B(a) - p_S(a)$ decreases in a and $\lim_{a \rightarrow \infty} p_B(a) = \lim_{a \rightarrow \infty} p_S(a) = p(a) = 1/2$. Proposition 3 also continues to hold as stated in this setting if we again replace the limit $\mu \rightarrow 1$ with $a \rightarrow \infty$.

With regard to the returns to scale in market making, we cannot analytically compute T_n and T_{n+1}/T_n in this model. However, the numerical calculations displayed in Figure 11 show that this model exhibits the same comparative statics as those in our baseline model.

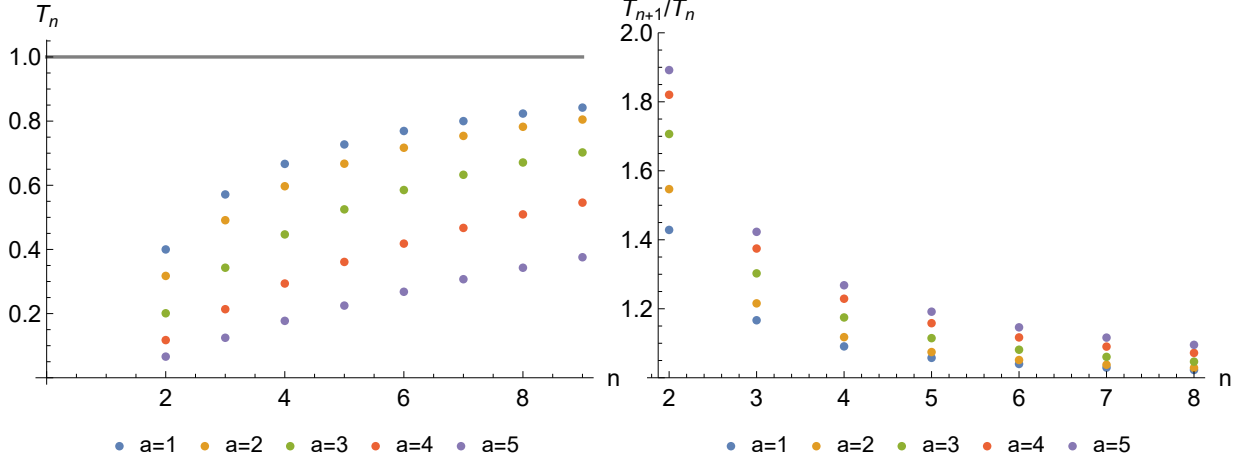


Figure 11: Illustration of our measure of market thickness T_n and returns to scale in market making T_{n+1}/T_n for uniform distributions and various values of a and n .

B.6 Competing exchanges: Second-best bilateral trade

In this section we reconsider the analysis from Section 4.2 assuming that the second-best mechanism is used in the competing bilateral exchanges. For this analysis, we restrict ourselves to the case in which F and G are uniform.

As before, we focus on the equilibrium with monotone sorting in which all buyers with $v \in (\bar{v}, 1]$ and all sellers with $c \in [0, \underline{c})$ trade via the monopoly and all the other types join the bilateral exchanges. We let $U_B(v)$ denote the expected utility of a buyer of type v from the bilateral exchange and $U_S(c)$ denote that of a seller of type c . Routine calculations show that the second-best mechanism has $\alpha^* = 1/3$ and is such that

$$U_B(v) = \frac{1-\mu}{32} (\bar{v}^2 - 2\bar{v}(\underline{c} + 4v) - 15\underline{c}^2 + 8\underline{c}v + 16v^2)$$

and

$$U_S(c) = \frac{1-\mu}{32} (16c^2 + 8c(\bar{v} - \underline{c} - 4) - 15\bar{v}^2 - 2\bar{v}(\underline{c} - 12) + \underline{c}(\underline{c} + 8)).$$

It again follows that $\bar{v}$ and $\underline{c}$ satisfy²⁵

$$\bar{v} - p_B = U_B(\bar{v}) \quad \text{and} \quad p_S - \underline{c} = U_S(\underline{c})$$

²⁵Note that since $U'_B(v) < 1$ and $U'_S(c) > -1$ the single crossing condition holds.

and the monopoly's spread is

$$p_B - p_S = \bar{v} - \underline{c} - (U_B(\bar{v}) + U_S(\underline{c})). \quad (15)$$

Again, noting that it is never optimal to choose p_B and p_S such that $1 - F(\bar{v}) \neq G(\underline{c})$, we can replace $1 - F(\bar{v})$ and $G(\underline{c})$ by the monopoly's "quantity" $q \in [0, 1]$. The monopolist's profit can then be written

$$\pi(q) = (1 - 2q - (U_B(\bar{v}) + U_S(\underline{c})))q = \frac{q}{16}(q - 2q)(9\mu - 6(1 - \mu)q + 7).$$

The corresponding first-order condition that pins down the optimal quantity q^* is then given by

$$7 + 36(q^*)^2(1 - \mu) + 9\mu - 8q^*(5 + 3\mu) = 0.$$

Solving this quadratic equation yields the two solutions

$$q^*(\mu) = \frac{10 + 6\mu + \sqrt{37 + 102\mu + 117\mu^2}}{18(1 - \mu)}, \quad \frac{10 + 6\mu - \sqrt{37 + 102\mu + 117\mu^2}}{18(1 - \mu)}.$$

Imposing the requirement that $q^*(\mu) \in [0, 1]$ then leaves us with

$$q^*(\mu) = \frac{10 + 6\mu - \sqrt{37 + 102\mu + 117\mu^2}}{18(1 - \mu)}.$$

Note that in this case $q^*(\mu)$ is increasing in μ . The spread of the monopolist is then given by

$$p_B - p_S = \frac{-99\mu^2 + 3\left(\sqrt{117\mu^2 + 102\mu + 37} - 14\right)\mu + 5\sqrt{117\mu^2 + 102\mu + 37} + 13}{216(1 - \mu)},$$

Note that as before the spread of the monopolist is increasing in μ . These comparative statics are illustrated in Figure 12.

B.7 Closed form solutions and illustration

For the special case of uniform distributions, we can provide closed-form expressions for social surplus, as well as our measures of market thickness and the returns to scale in market making.

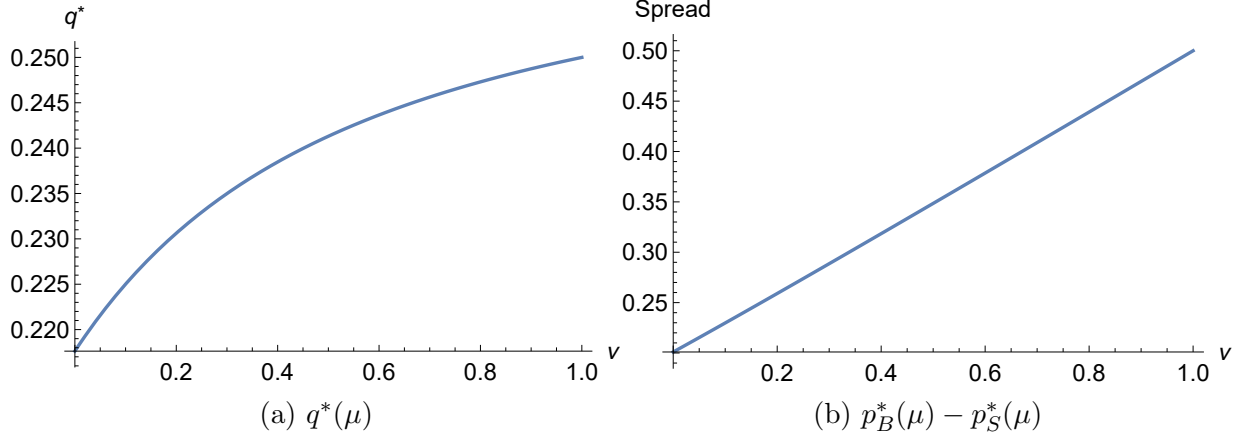


Figure 12: Equilibrium quantity $q^*(\mu)$ (panel (a)) and the equilibrium spread $p_B^*(\mu) - p_S^*(\mu)$ (panel (b)) as a function of μ for F and G uniform.

Corollary B.1. *If F and G are uniform distributions, then we have*

$$S_n^{FB}(\mu) = \frac{n}{(4n+2)(2n-1)} \left(n(2n-1) + (n(2n+1)-1)\mu + \sum_{i=2}^{2n} \mu^i \right).$$

Consequently, we have

$$T_n(\mu) = \frac{2(n-1)(2n-1) - 6(n-1)(2n+1)\mu^2 + 2(4n^2-1)\mu^3 - 6\mu^{2n+1}}{(4n^2-1)(2\mu+1)(1-\mu)^2}$$

and

$$\frac{T_{n+1}(\mu)}{T_n(\mu)} = \frac{(2n-1)[2(1-\mu)^2(2\mu+1)n^2 + 3\mu^3(1-\mu^{2n}) + n(8\mu^3 - 9\mu^2 + 1)]}{(2n+3)[1 + 3n(\mu^2-1) + 3\mu^2 - \mu^3 + 2n^2(\mu-1)^2(2\mu+1) - 3\mu^{2n+1}]}$$

For mass products (i.e. setting $\mu = 0$), this yields

$$S_n^{FB}(0) = \frac{n^2}{4n+2}, \quad T_n(0) = \frac{2(n-1)}{2n+1} \quad \text{and} \quad \frac{T_{n+1}(0)}{T_n(0)} = \frac{n(2n+1)}{2n^2+n-3}.$$

Proof. Adopting the notation from the proof of Proposition 5 and combining

$$\begin{aligned} h_{\mu,n,n,n}(x) &= \sum_{i=0}^{n-1} (n-i) \binom{n}{i} \binom{n}{i} f_{\mu}(x) F_{\mu}(x)^i (1-G_{\mu}(x))^i (1-F_{\mu}(x))^{n-i-1} G_{\mu}(x)^{n-i} \\ &\quad + \sum_{i=0}^{n-1} (n-i) \binom{n}{i} \binom{n}{i} g_{\mu}(x) (1-F_{\mu}(x))^i G_{\mu}(x)^i F_{\mu}(x)^{n-i} (1-G_{\mu}(x))^{-i+n-1} \end{aligned}$$

and

$$\begin{aligned}
& h_{\mu, n+1, n, n+1}(x) \\
&= \sum_{i=0}^n (n-i+1) \binom{n+1}{i} \binom{n}{i} f_{\mu}(x) F_{\mu}(x)^i (1-G_{\mu}(x))^i (1-F_{\mu}(x))^{n-i} G_{\mu}(x)^{n-i} \\
&+ \sum_{i=0}^{n-1} (n-i) \binom{n}{i} \binom{n+1}{i+1} g_{\mu}(x) (1-F_{\mu}(x))^{i+1} G_{\mu}(x)^i F_{\mu}(x)^{n-i} (1-G_{\mu}(x))^{-i+n-1}.
\end{aligned}$$

with the results of Proposition 5 yields the desired expressions. □